

Visualisation Examples Catalogue

Decision-support toolkit for visualising accelerometer-derived movement behaviour data

32 generated visualisation examples: 9 application-to-a-published-study examples and 23 simulated data examples.

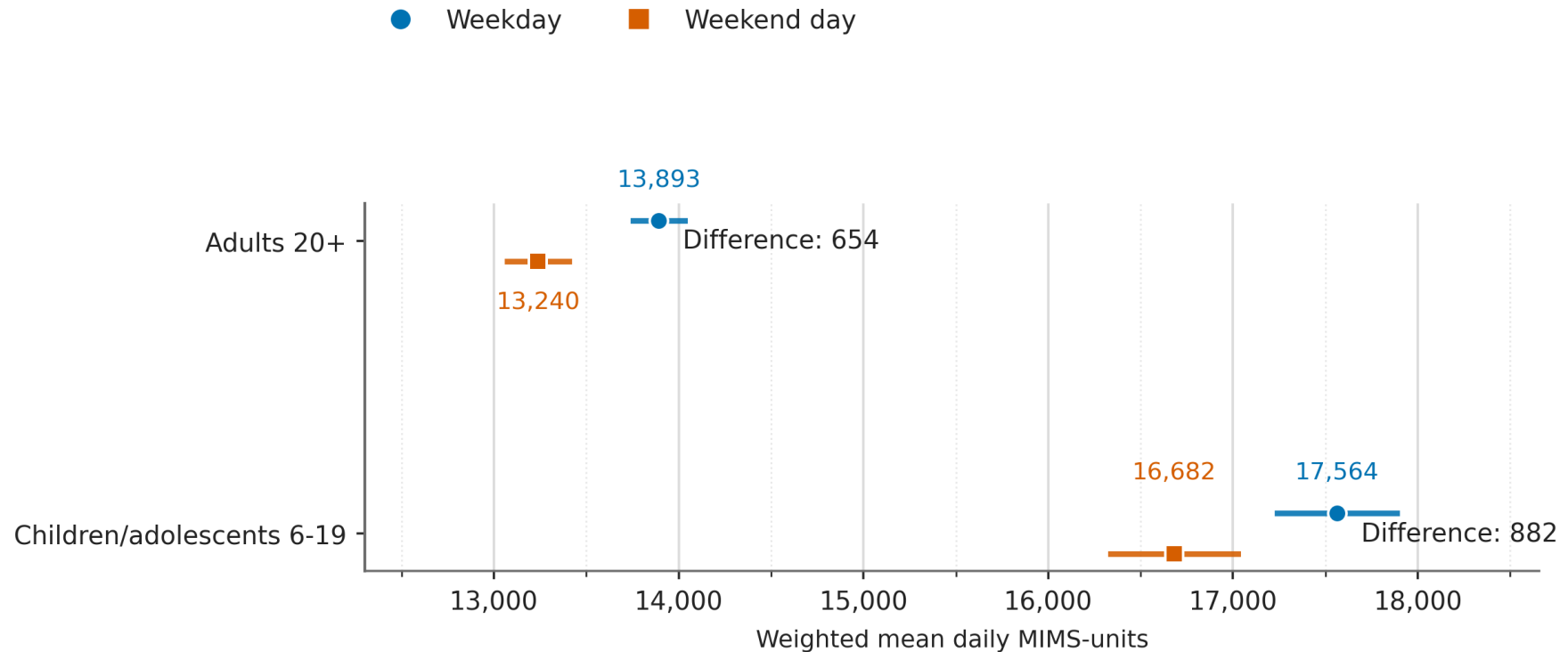
Application to a published study examples were generated from the reproduced NHANES 2011-2014 outputs reconstructed for the published-study application. Simulated data examples were generated from small mock datasets to illustrate recommendation-specific visual structures and should not be interpreted as evidence that the designs have been empirically tested.

Application to a published study examples

Figures generated from the reconstructed NHANES dataset and associated summary outputs used in the published-study application.

Published-study application example 1. Weekday and weekend-day physical activity by sample

Weekday and weekend-day physical activity by sample



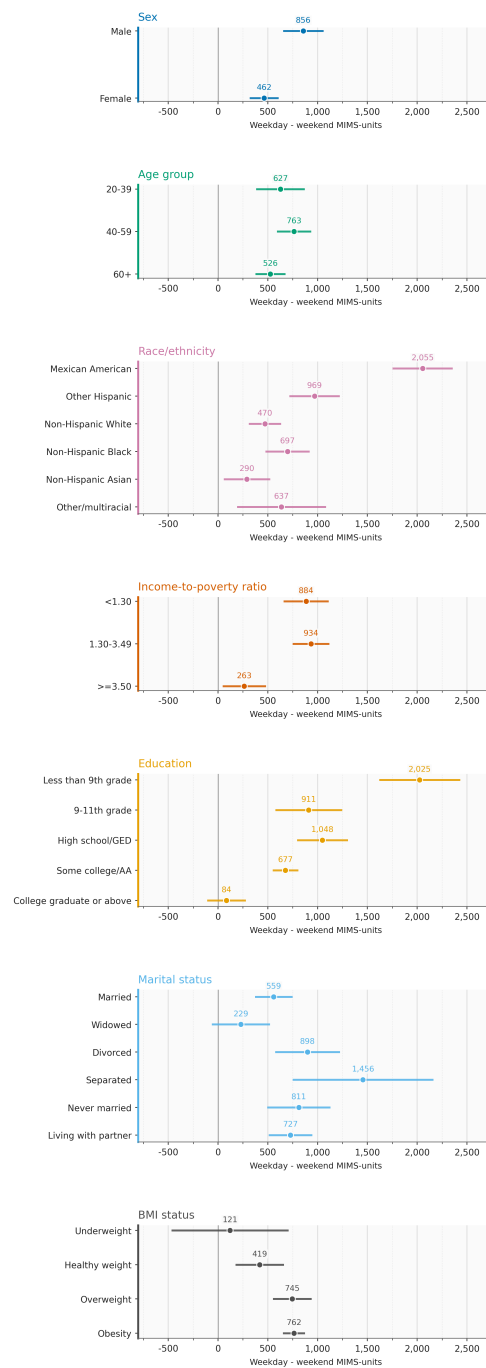
Points show weighted estimates; horizontal intervals show 95% confidence intervals. MIMS = Monitor-Independent Movement Summary units. The x-axis is zoomed to make the paired day-type estimates readable and starts at 12,302 MIMS-units, not zero; values are shown in the original metric units.

Weighted mean daily MIMS-units on weekdays and weekend days for adults and children/adolescents. Points show estimates, horizontal intervals show 95% confidence intervals, and text labels report the descriptive weekday-minus-weekend difference. No statistical significance marker is shown.

Source: Application to a published study, reproduced NHANES 2011-2014 data.

Published-study application example 2. Weekday-weekend physical activity difference by subgroup: Adults 20+

Weekday-weekend physical activity difference by subgroup: Adults 20+



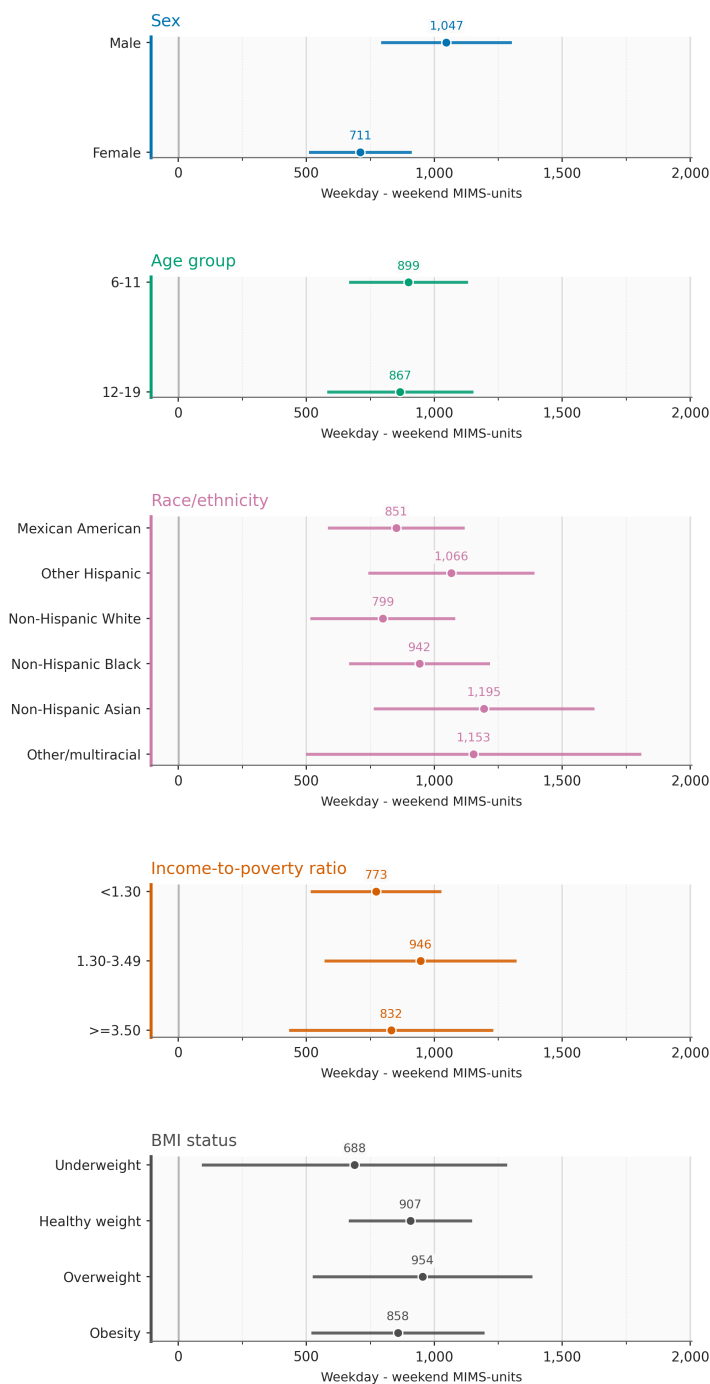
Points show weighted estimates; horizontal intervals show 95% confidence intervals. MIM5 = Monitor-Independent Movement Summary units. Positive values indicate higher weekday activity. Panel colours identify subgroup domains.

Weighted weekday-minus-weekend MIM5-unit differences by subgroup for Adults 20+. Positive values indicate higher weekday physical activity. Points show estimates and horizontal intervals show 95% confidence intervals. Panel colours identify subgroup domains.

Source: Application to a published study, reproduced NHANES 2011-2014 data.

Published-study application example 3. Weekday-weekend physical activity difference by subgroup: Children/adolescents 6-19

Weekday-weekend physical activity difference by subgroup: Children/adolescents 6-19



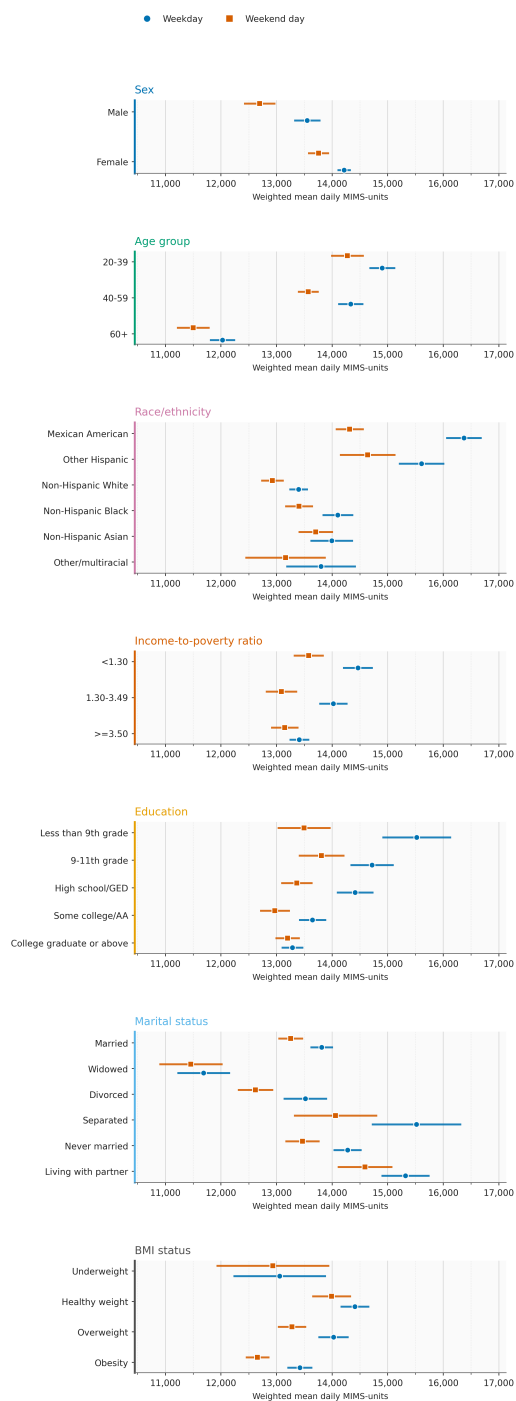
Points show weighted estimates; horizontal intervals show 95% confidence intervals. MIMS = Monitor-Independent Movement Summary units. Positive values indicate higher weekday activity. Panel colours identify subgroup domains.

Weighted weekday-minus-weekend MIMS-unit differences by subgroup for Children/adolescents 6-19. Positive values indicate higher weekday physical activity. Points show estimates and horizontal intervals show 95% confidence intervals. Panel colours identify subgroup domains.

Source: Application to a published study, reproduced NHANES 2011-2014 data.

Published-study application example 4. Weekday and weekend-day physical activity by subgroup: Adults 20+

Weekday and weekend-day physical activity by subgroup: Adults 20+



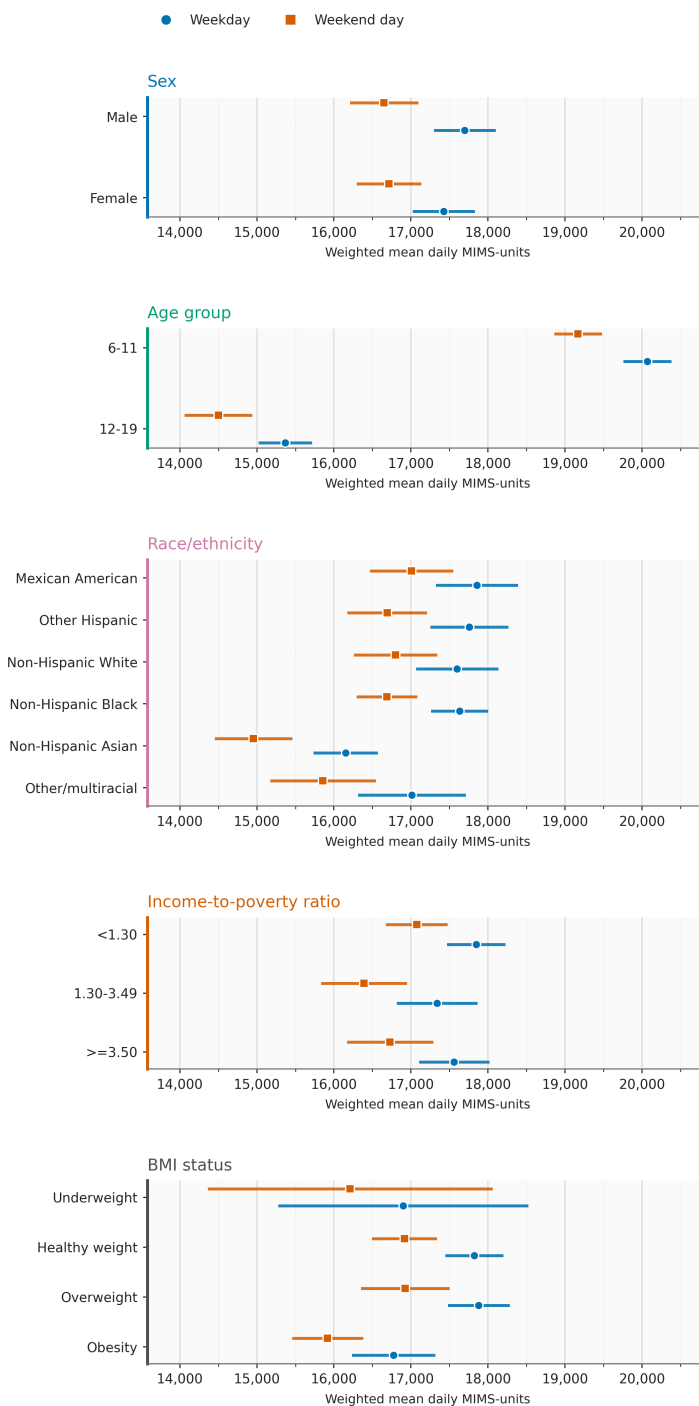
Points show weighted estimates; horizontal intervals show 95% confidence intervals. MIMS = Monitor-Independent Movement Summary units. The x-axis is scaled to show subgroup comparisons; dotted minor gridlines help locate values. Panel colours identify subgroup domains.

Weighted mean daily MIMS-units on weekdays and weekend days by subgroup for Adults 20+. Points show estimates and horizontal intervals show 95% confidence intervals. Panel colours identify subgroup domains.

Source: Application to a published study, reproduced NHANES 2011-2014 data.

Published-study application example 5. Weekday and weekend-day physical activity by subgroup: Children/adolescents 6-19

Weekday and weekend-day physical activity by subgroup: Children/adolescents 6-19



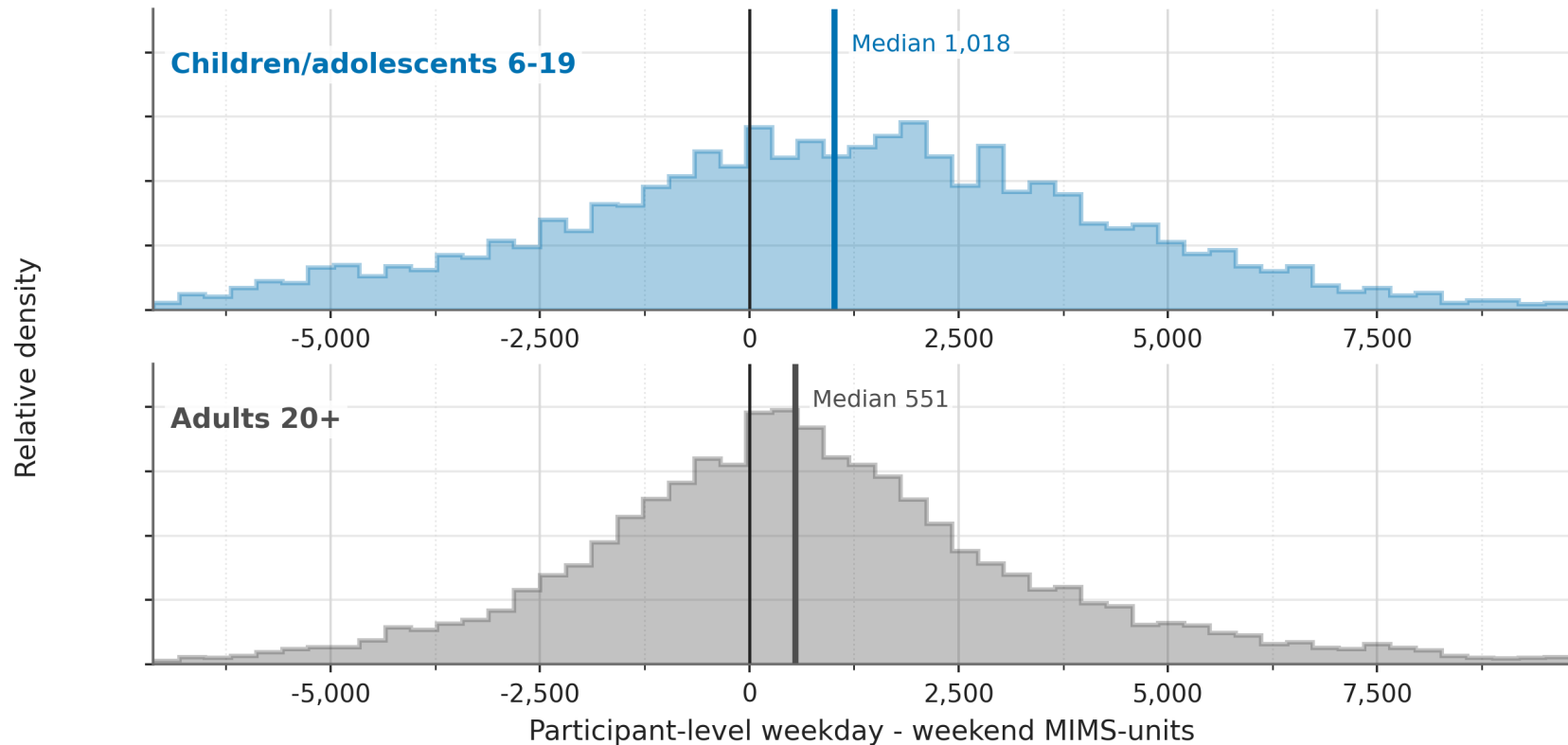
Points show weighted estimates; horizontal intervals show 95% confidence intervals. MIMS = Monitor-Independent Movement Summary units. The x-axis is scaled to show subgroup comparisons; dotted minor gridlines help locate values. Panel colours identify subgroup domains.

Weighted mean daily MIMS-units on weekdays and weekend days by subgroup for Children/adolescents 6-19. Points show estimates and horizontal intervals show 95% confidence intervals. Panel colours identify subgroup domains.

Source: Application to a published study, reproduced NHANES 2011-2014 data.

Published-study application example 6. Distribution of individual weekday-weekend differences

Distribution of individual weekday-weekend differences



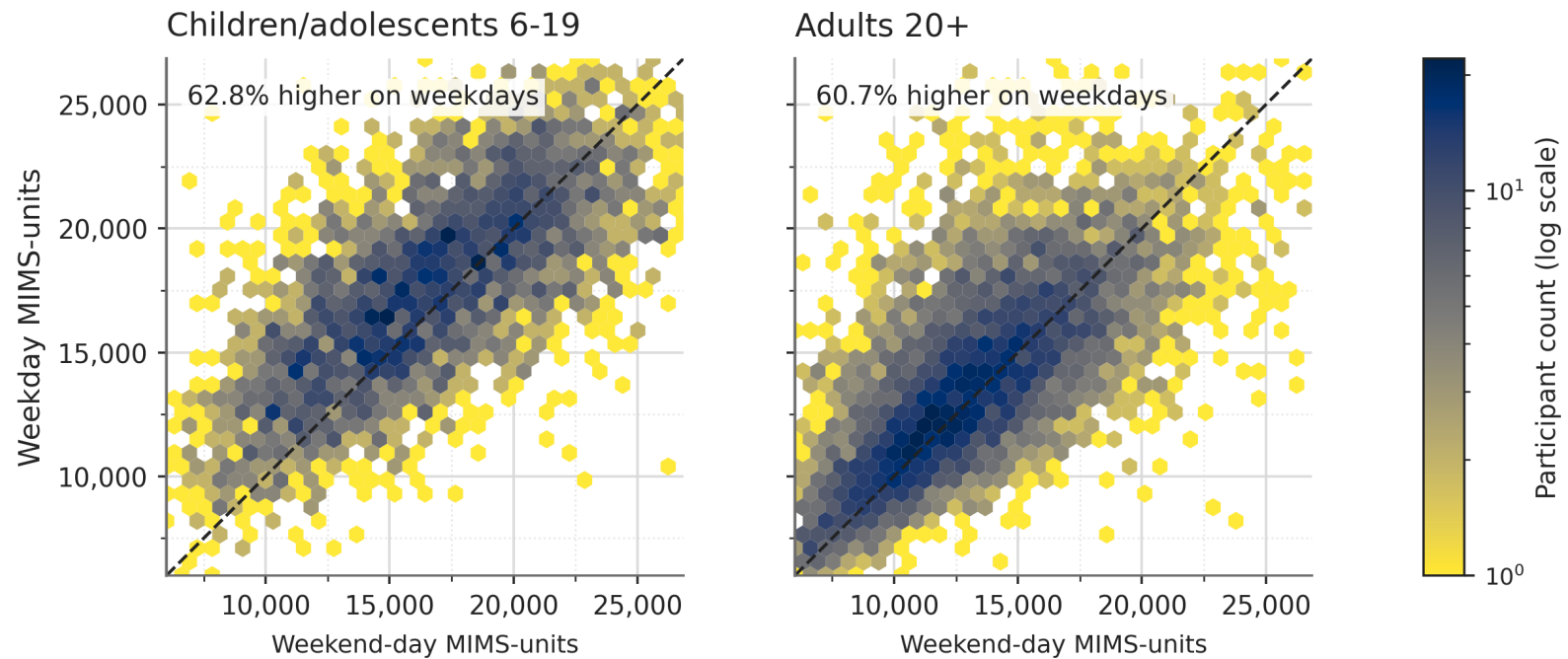
Distributions use unweighted participant-level differences and show the central 98% of values to keep the distribution readable; median lines are calculated from all retained participants. Positive values indicate higher weekday activity.

Distribution of participant-level weekday-minus-weekend MIMS-unit differences for adults and children/adolescents. Positive values indicate higher weekday physical activity. The axis shows the central 98% of values for readability.

Source: Application to a published study, reproduced NHANES 2011-2014 data.

Published-study application example 7. Participant-level relationship between weekday and weekend-day activity

Participant-level relationship between weekday and weekend-day activity



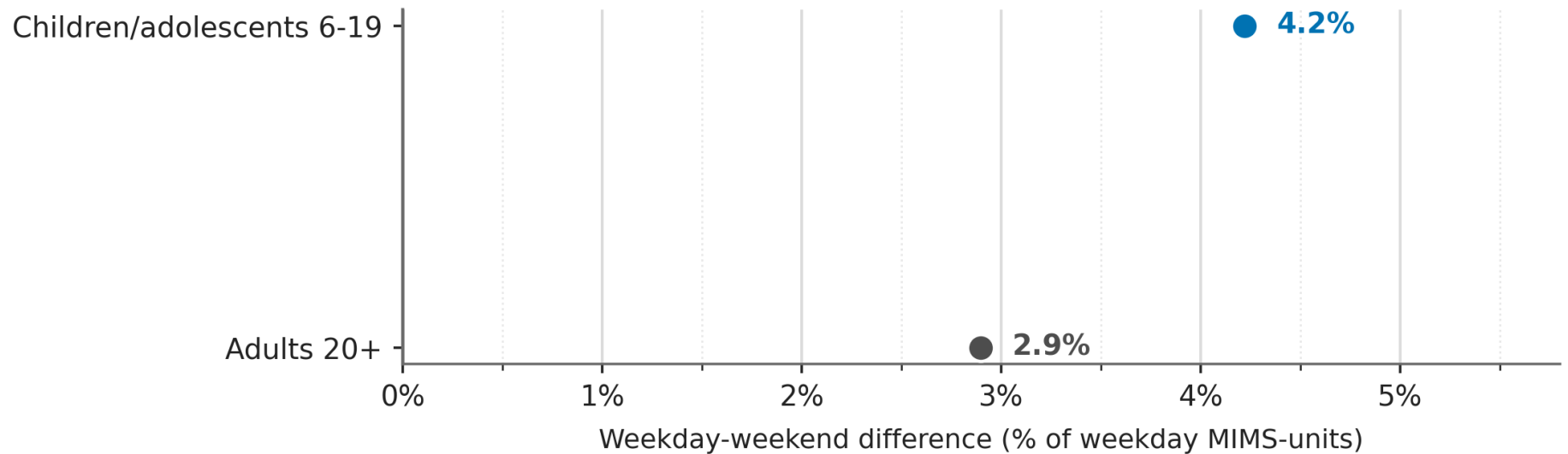
Each hexagon represents participants with similar weekday and weekend-day MIMS values; darker colour means more participants. The dashed line marks equal weekday and weekend-day activity. Axes show the central 98% of participant values for readability.

Hexbin plots showing the participant-level relationship between mean weekend-day and weekday MIMS-units. The dashed line represents equal weekday and weekend-day activity; points above the line indicate higher weekday activity.

Source: Application to a published study, reproduced NHANES 2011-2014 data.

Published-study application example 8. Percentage weekday-weekend difference by sample

Percentage weekday-weekend difference by sample



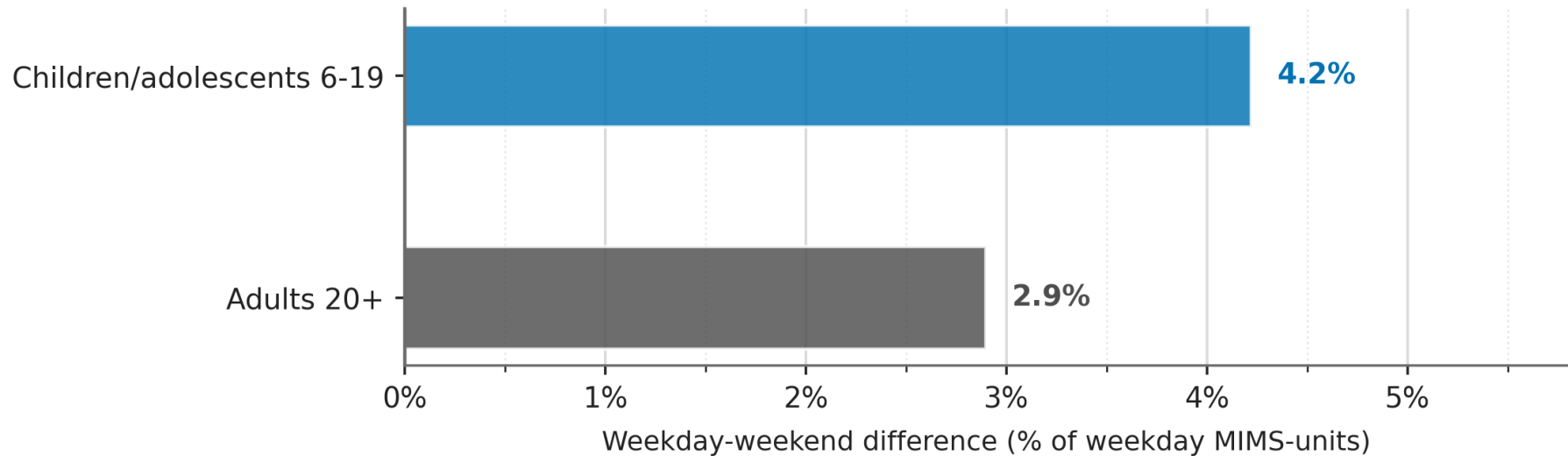
Dots show reconstructed weighted percentage differences. Intervals are not drawn in this example because the percentage column is treated as secondary descriptive context; use a point-range plot when uncertainty is central to the message.

Reconstructed weekday-weekend percentage differences relative to weekday MIMS-units for adults and children/adolescents. Dots show weighted summary percentages without intervals.

Source: Application to a published study, reproduced NHANES 2011-2014 data.

Published-study application example 9. Conditional bar-chart example: percentage difference by sample

Conditional bar-chart example: percentage difference by sample



This bar version is a conditional alternative because the percentage difference has a meaningful zero and the message is magnitude from zero. Prefer point-range or dot plots when uncertainty, distribution, or precise comparisons are the main message.

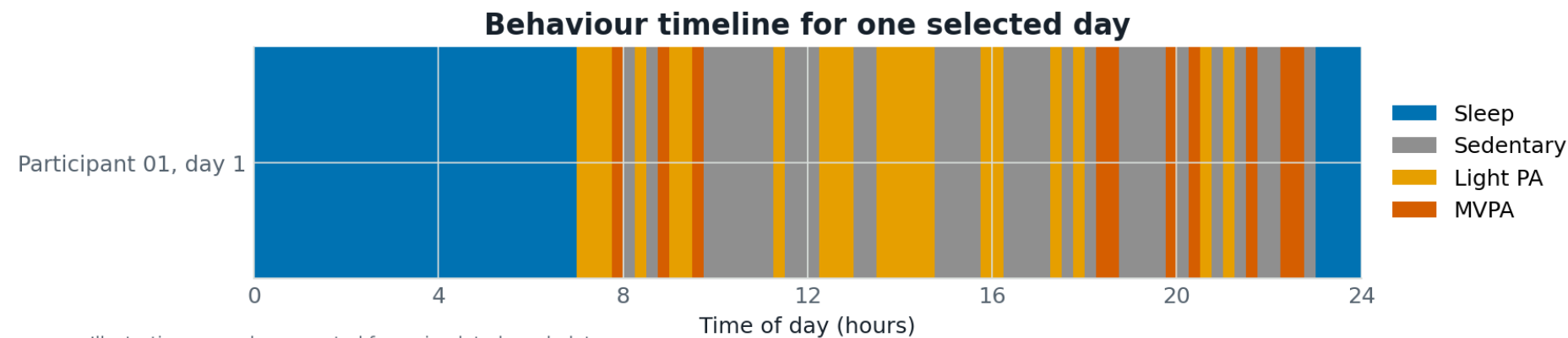
Conditional bar-chart version of the reconstructed weekday-weekend percentage differences relative to weekday MIMS-units. Bar length represents percentage difference from zero.

Source: Application to a published study, reproduced NHANES 2011-2014 data.

Simulated data examples

Figures generated from small simulated mock datasets to illustrate recommendation-specific visual structures.

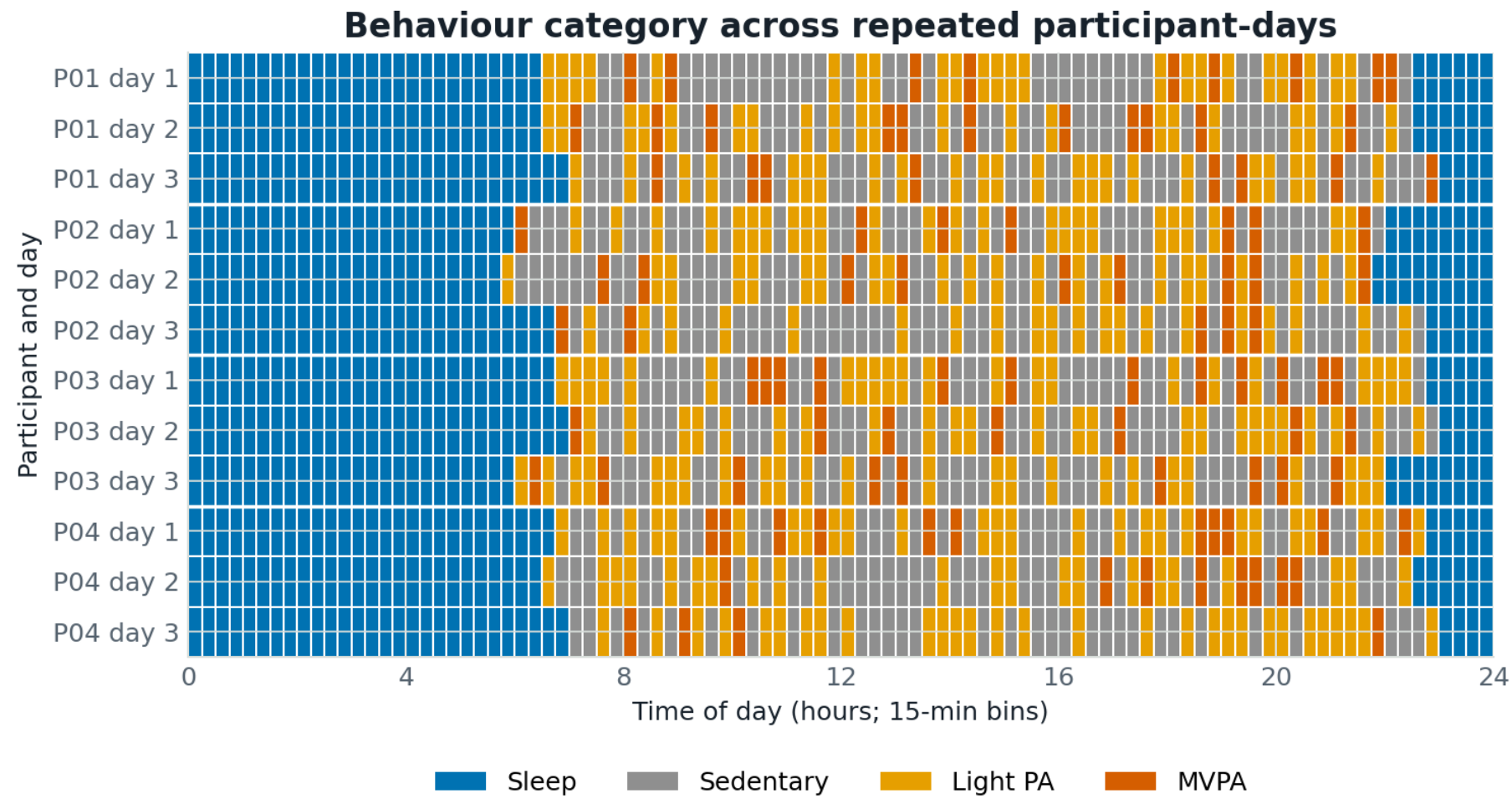
Simulated example 1. Behaviour timeline (tile plot)



Simulated data example illustrating the visual structure for the recommendation: Behaviour timeline (tile plot).

Source: Simulated mock data.

Simulated example 2. Behaviour-by-time heatmap

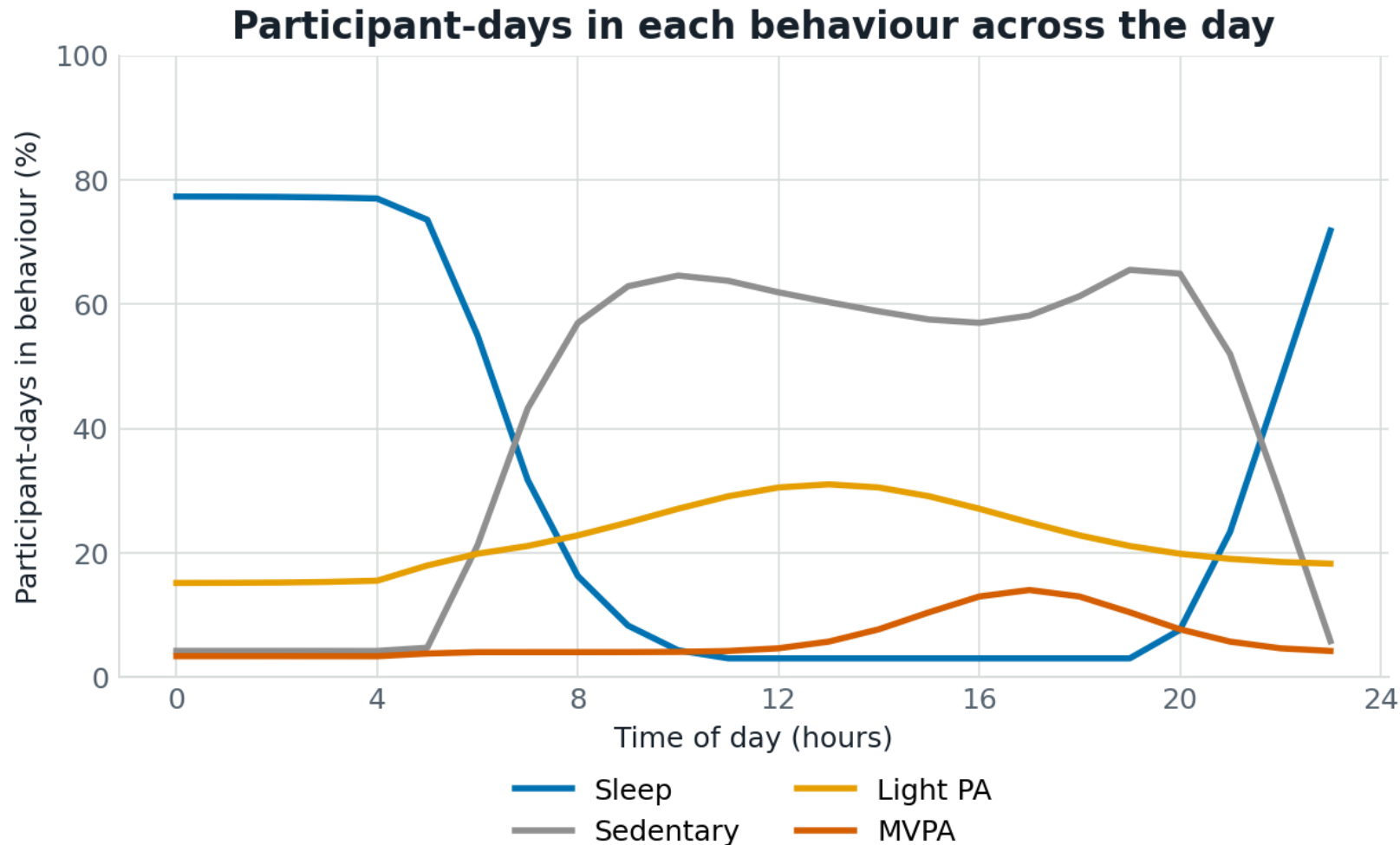


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Behaviour-by-time heatmap.

Source: Simulated mock data.

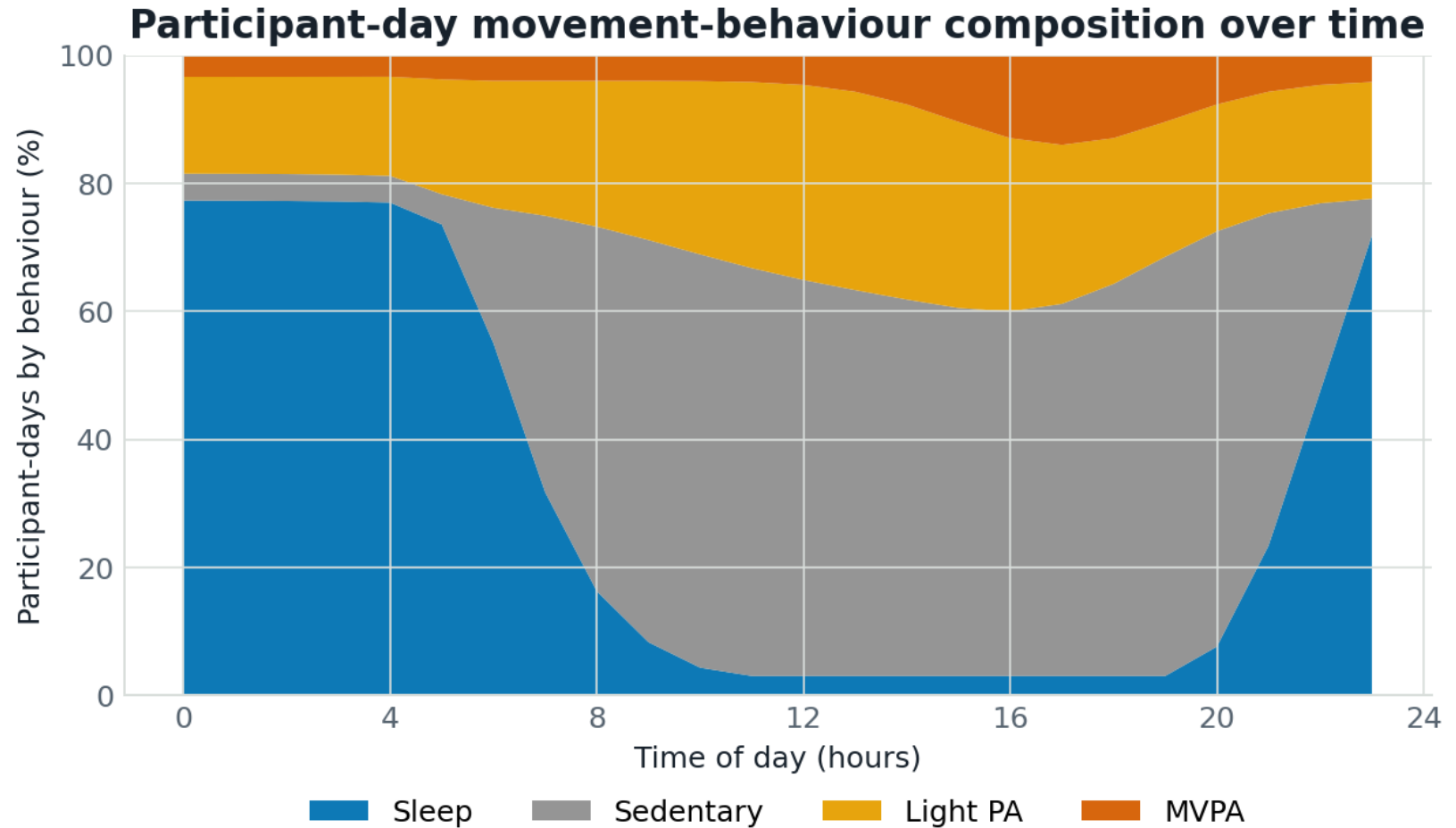
Simulated example 3. Proportion-over-time profile



Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Proportion-over-time profile.
Source: Simulated mock data.

Simulated example 4. Stacked area profile

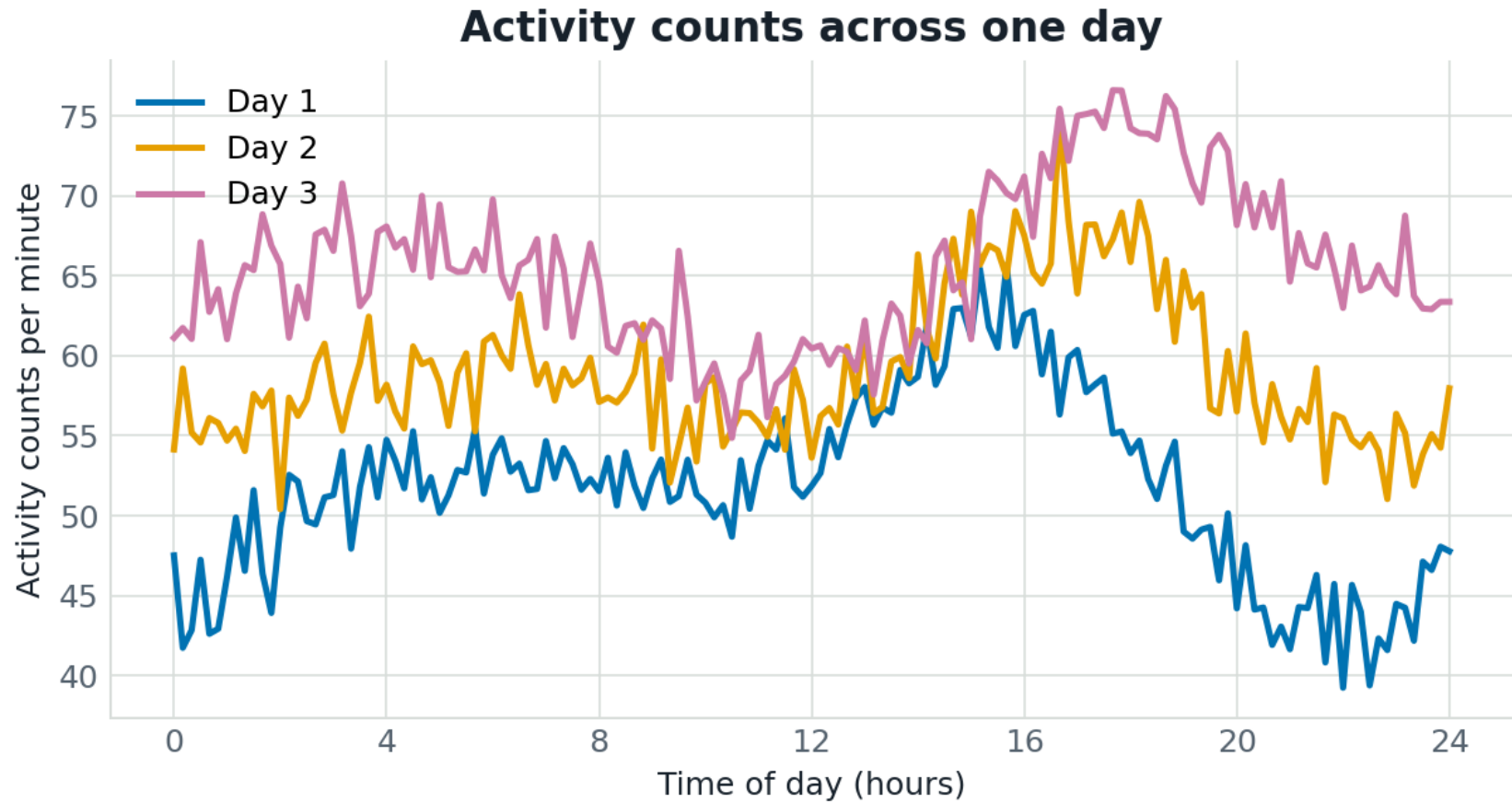


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Stacked area profile.

Source: Simulated mock data.

Simulated example 5. Time-series line plot

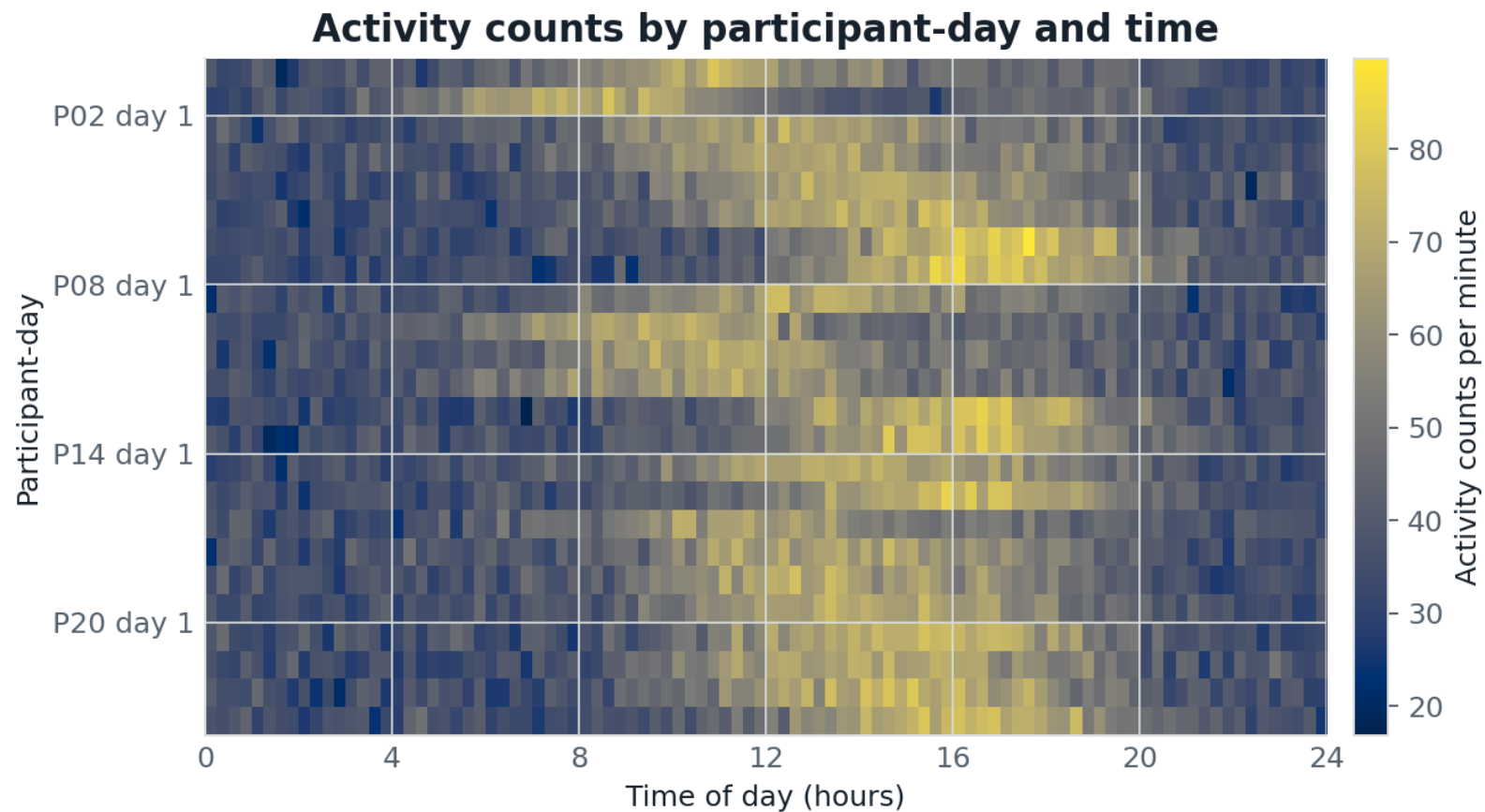


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Time-series line plot.

Source: Simulated mock data.

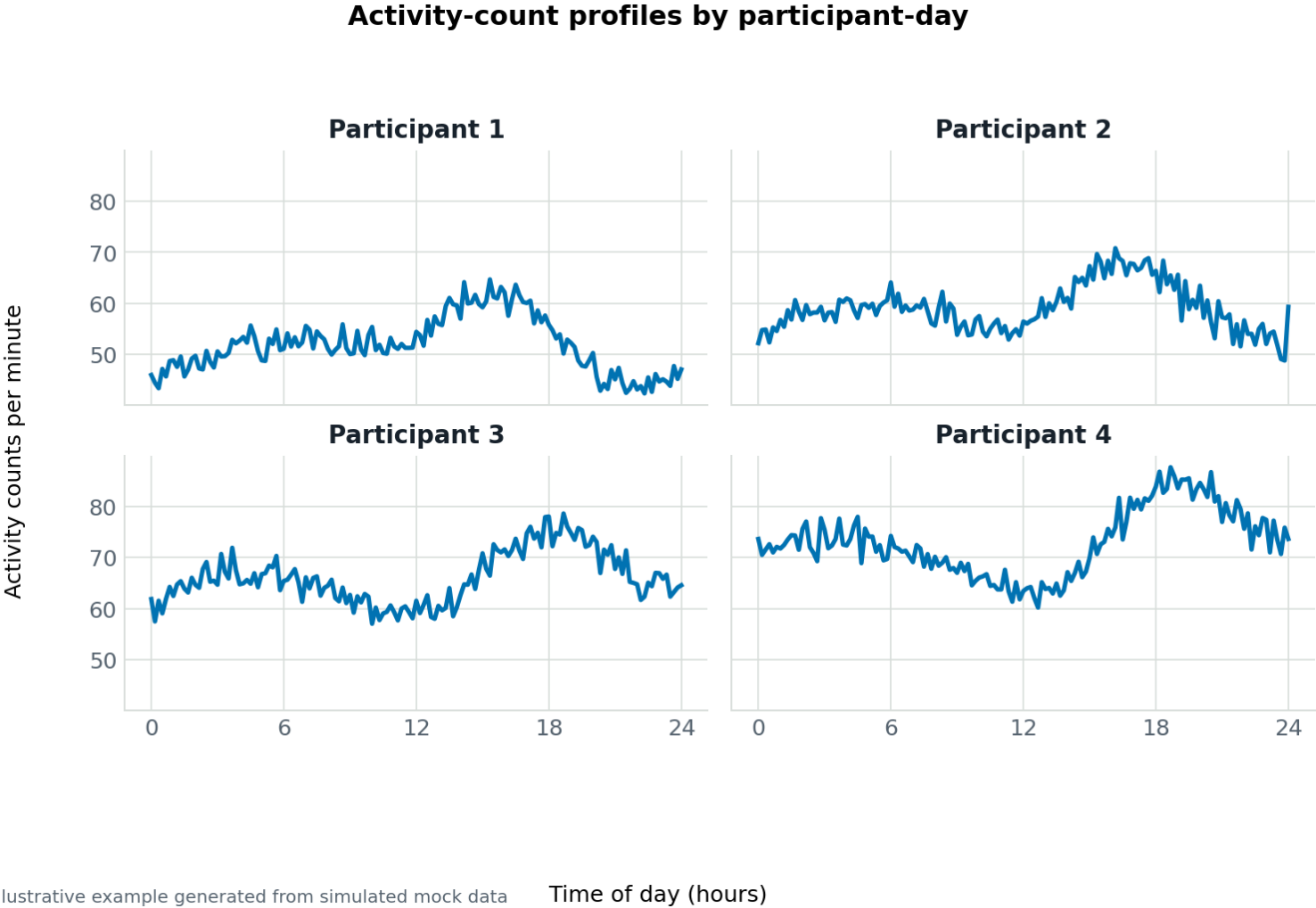
Simulated example 6. Observation-by-time heatmap



Illustrative example generated from simulated mock data

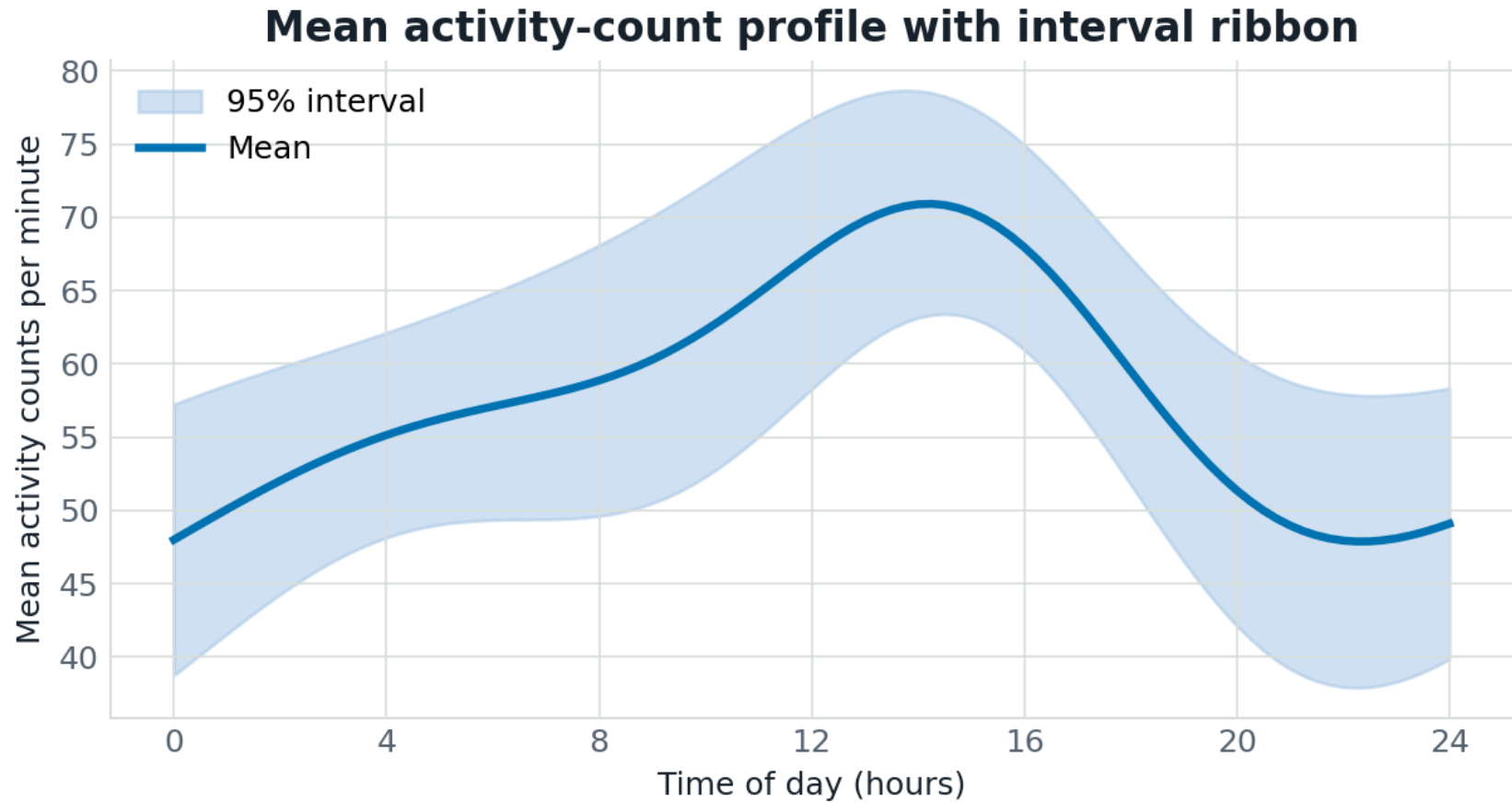
Simulated data example illustrating the visual structure for the recommendation: Observation-by-time heatmap.
Source: Simulated mock data.

Simulated example 7. Small-multiple time-series plots



Simulated data example illustrating the visual structure for the recommendation: Small-multiple time-series plots.
Source: Simulated mock data.

Simulated example 8. Summary time profile with interval ribbon

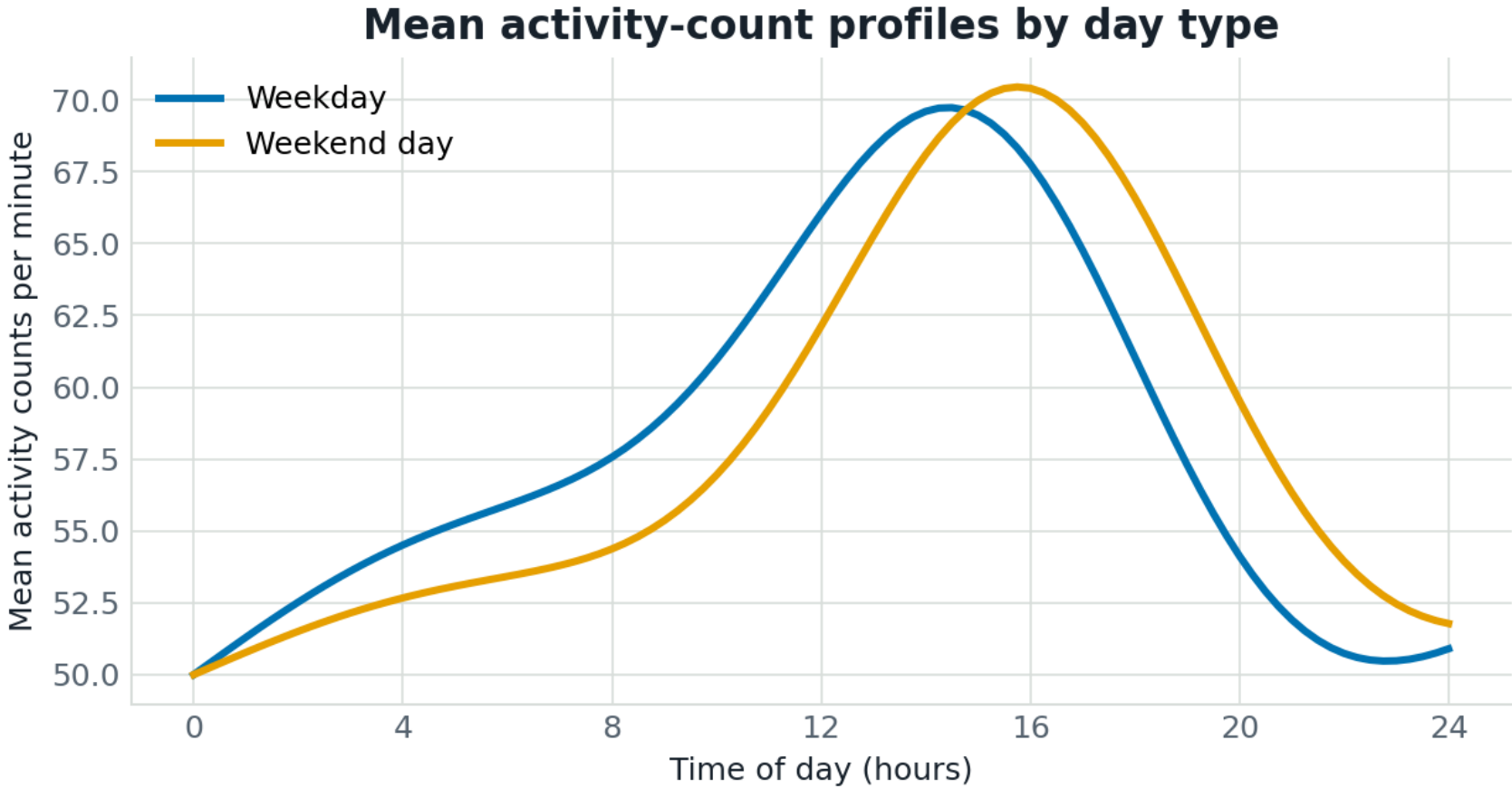


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Summary time profile with interval ribbon.

Source: Simulated mock data.

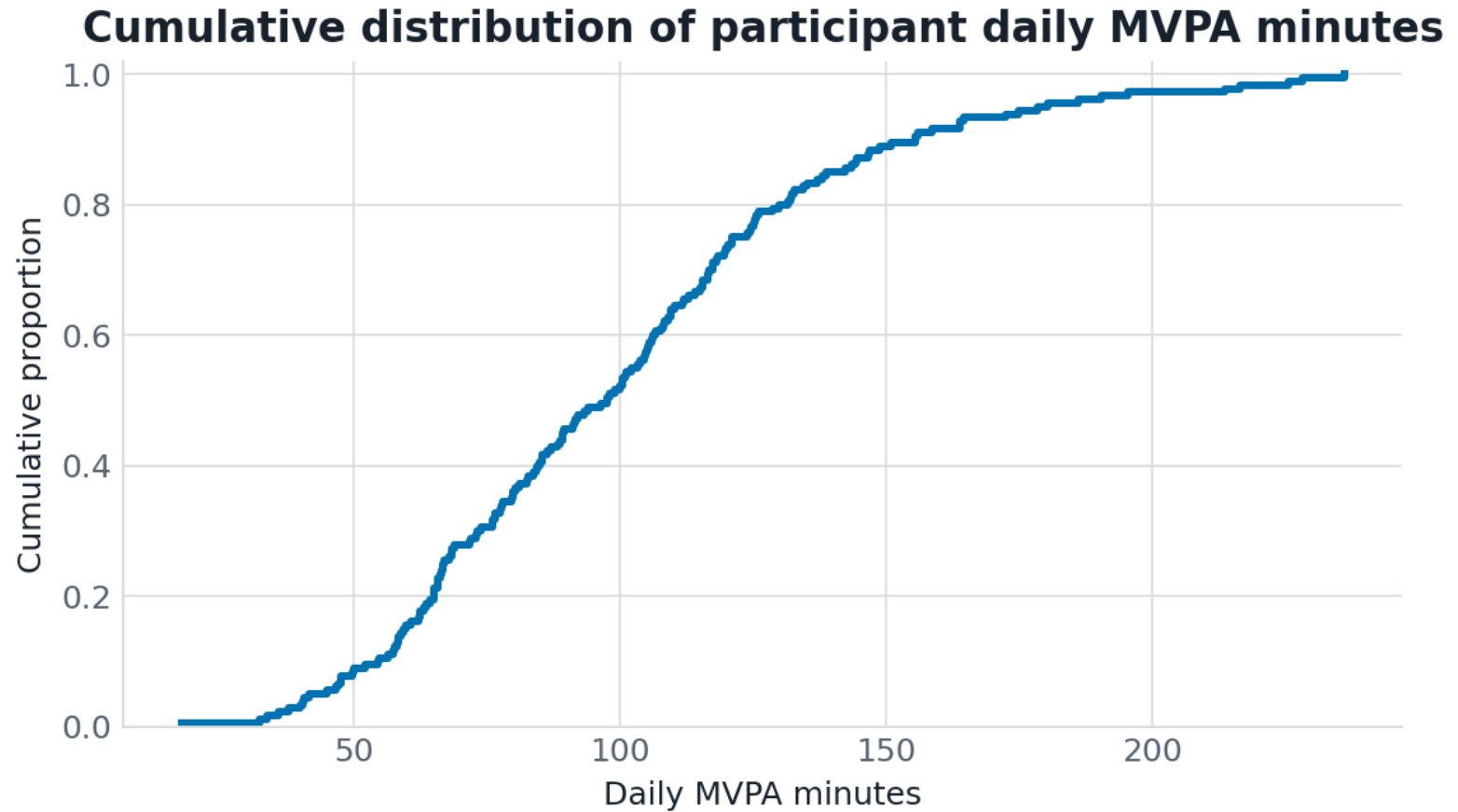
Simulated example 9. Summary time profile



Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Summary time profile.
Source: Simulated mock data.

Simulated example 10. Empirical cumulative distribution (ECDF)

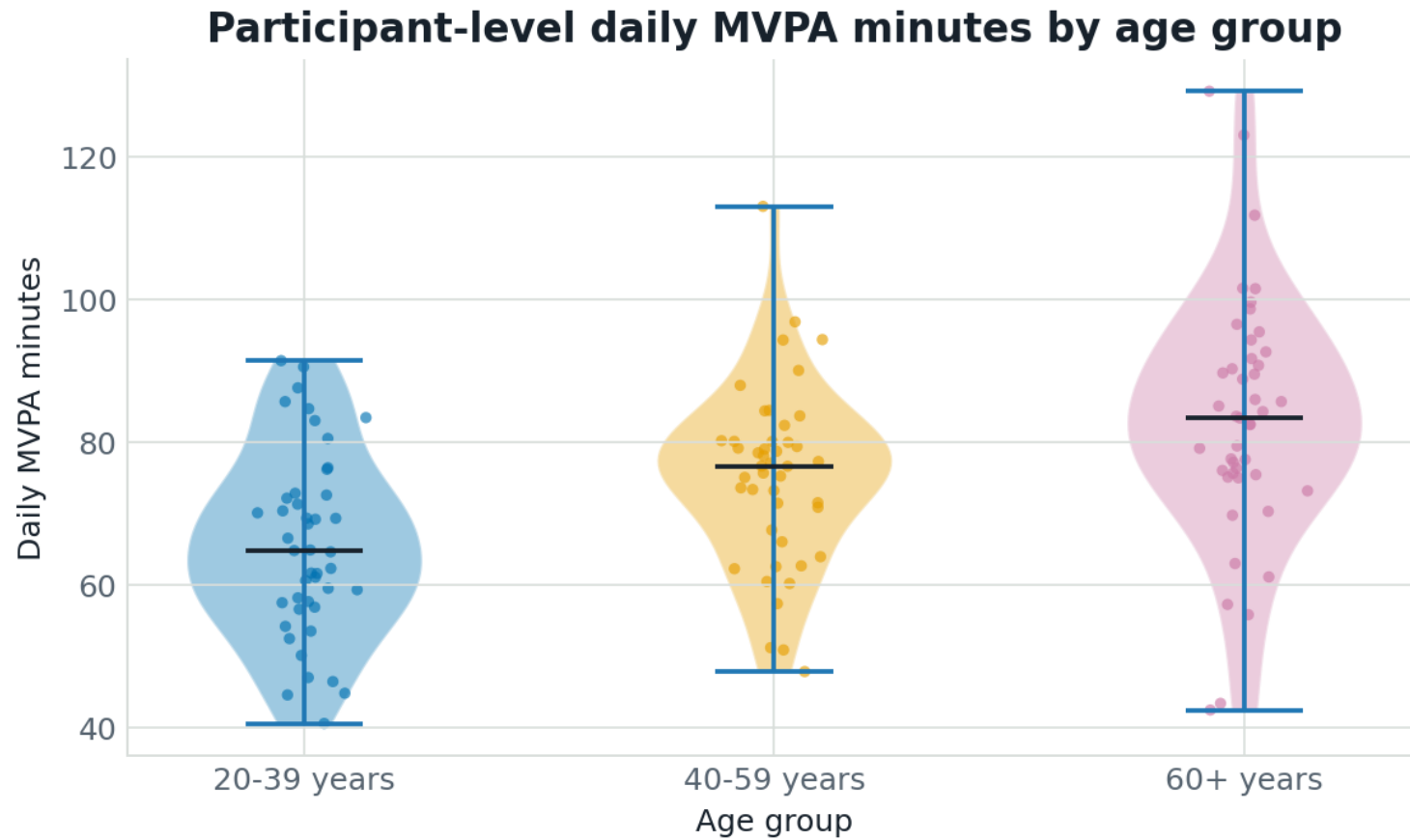


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Empirical cumulative distribution (ECDF).

Source: Simulated mock data.

Simulated example 11. Box or violin plot with raw points

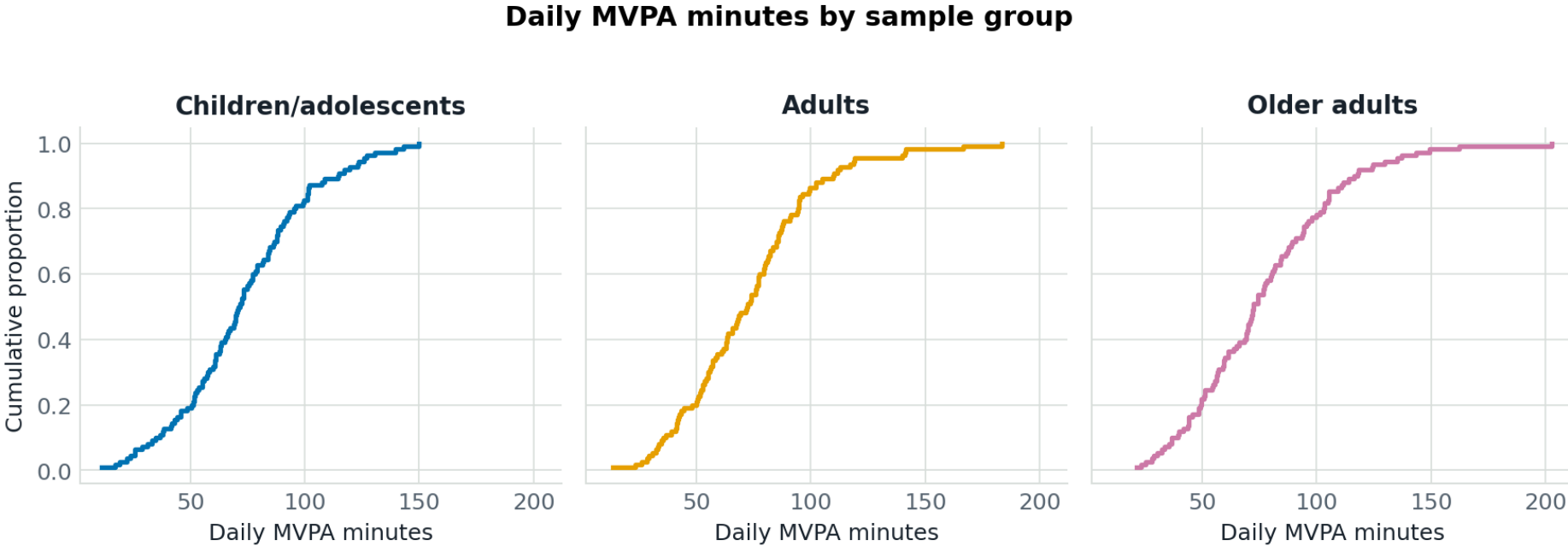


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Box or violin plot with raw points.

Source: Simulated mock data.

Simulated example 12. Faceted density or ECDF plot

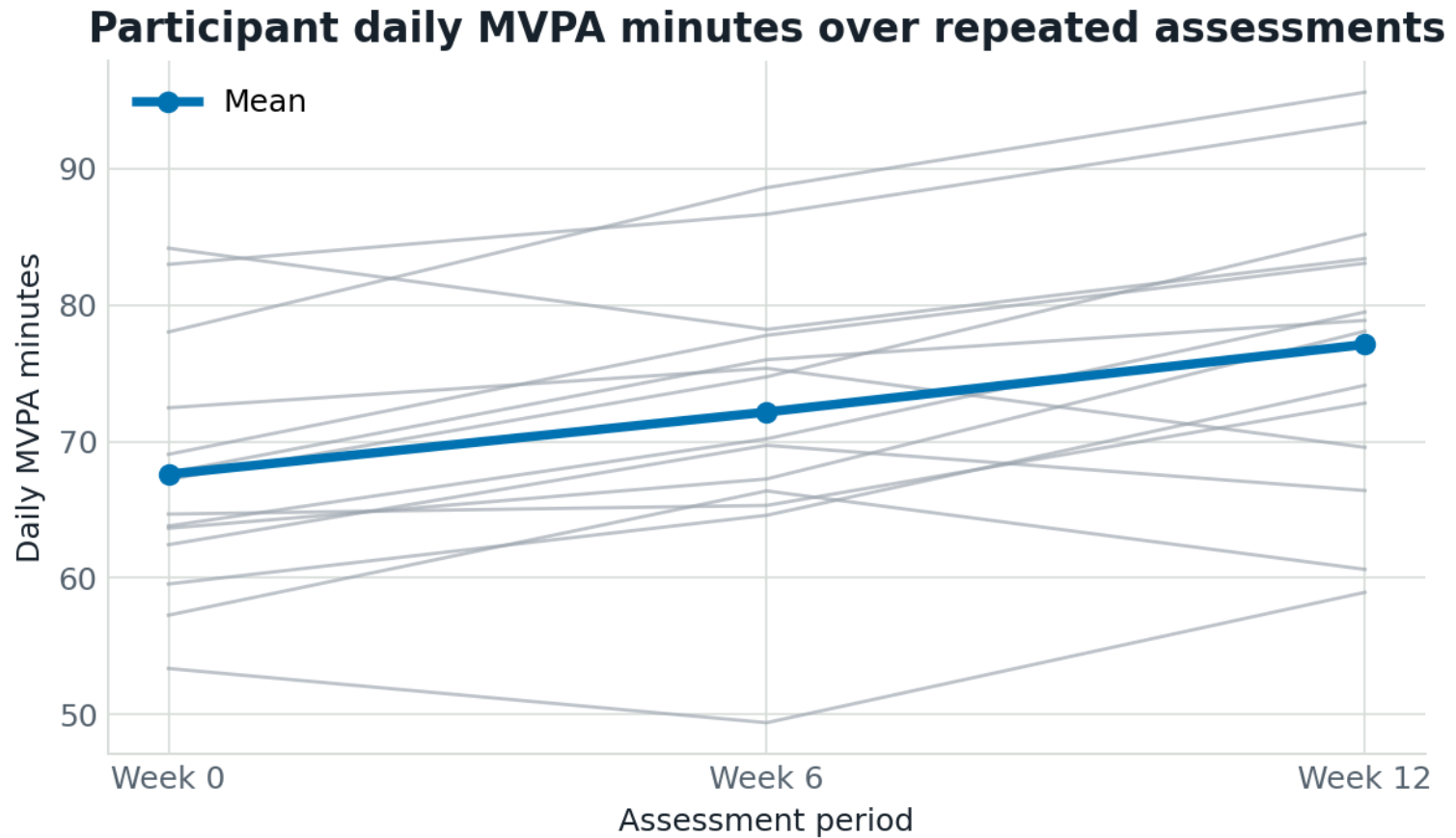


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Faceted density or ECDF plot.

Source: *Simulated mock data.*

Simulated example 13. Repeated-measures line plot

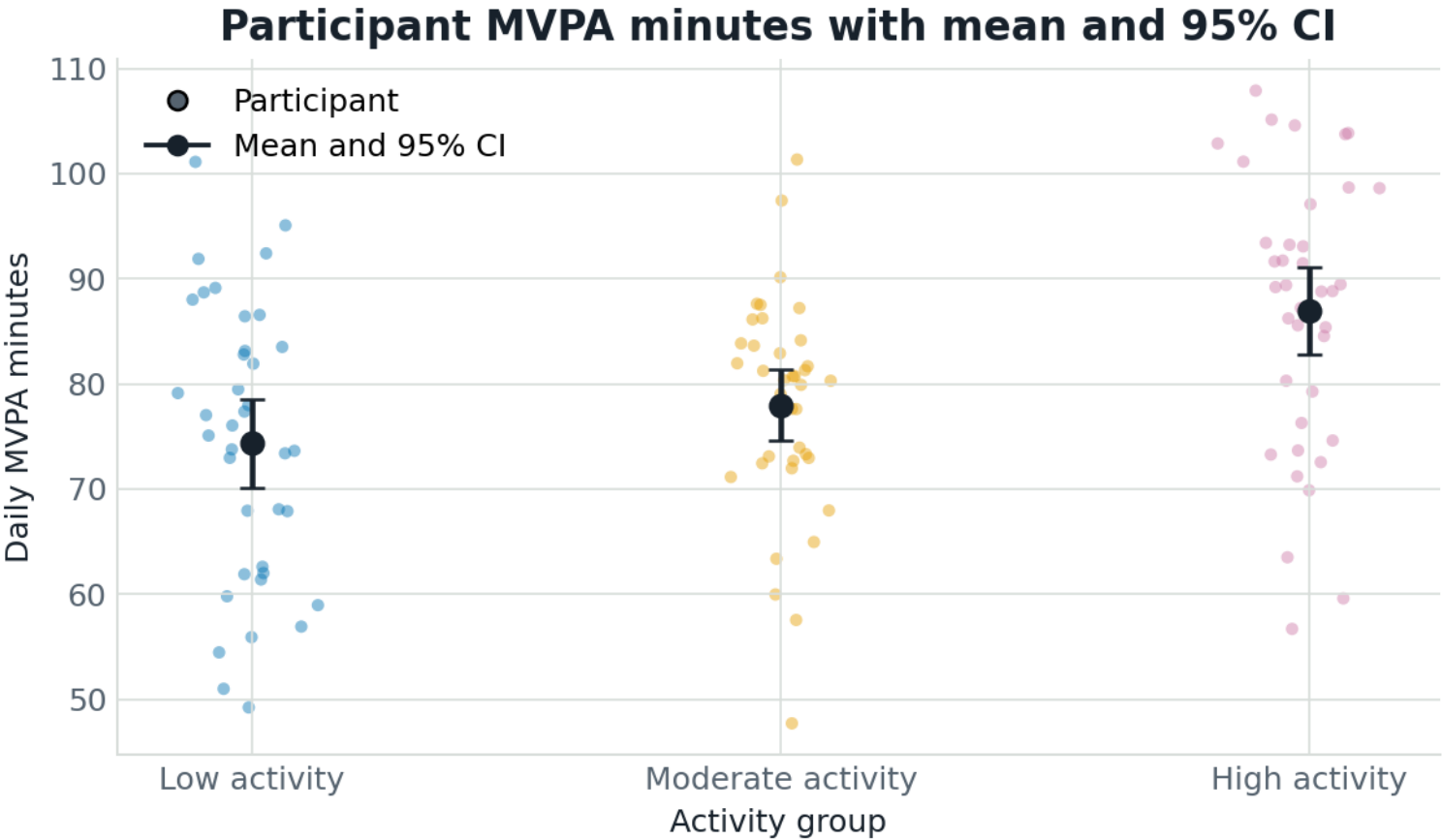


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Repeated-measures line plot.

Source: Simulated mock data.

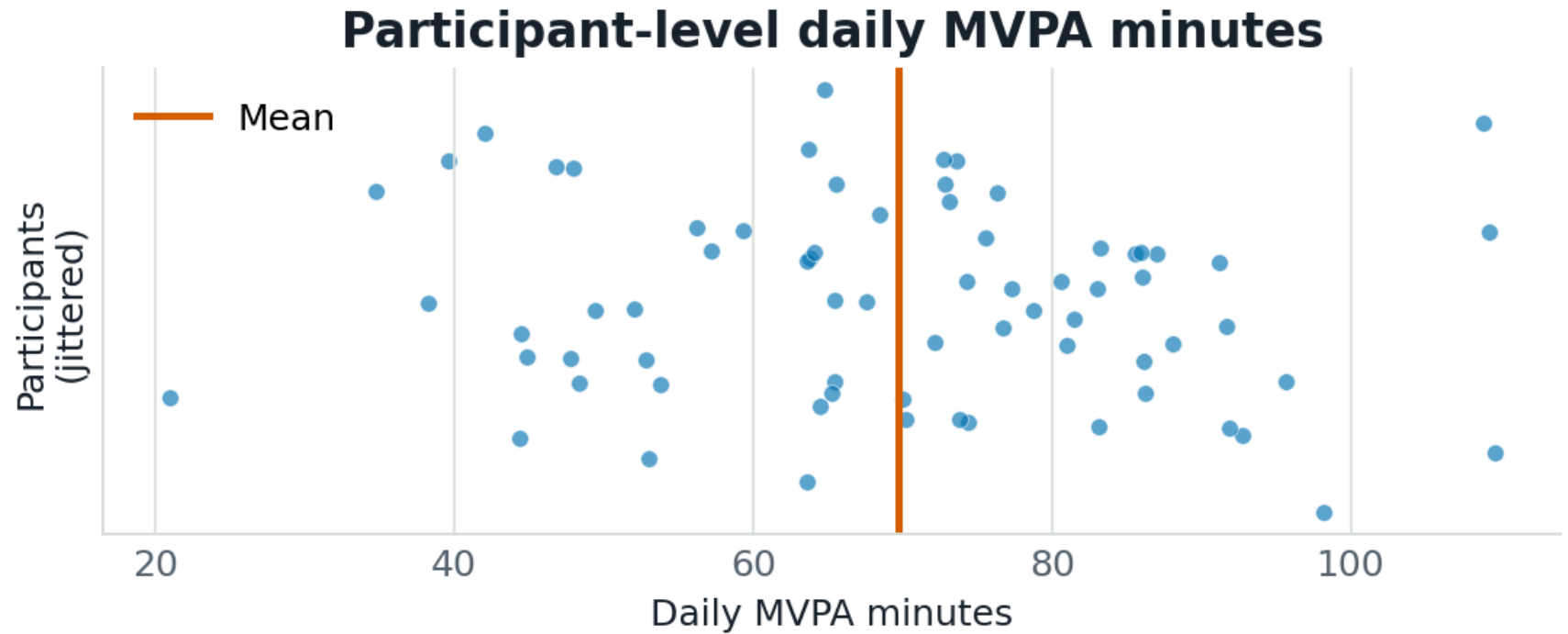
Simulated example 14. Dot plot with summary and interval



Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Dot plot with summary and interval.
Source: Simulated mock data.

Simulated example 15. Dot plot of observed values

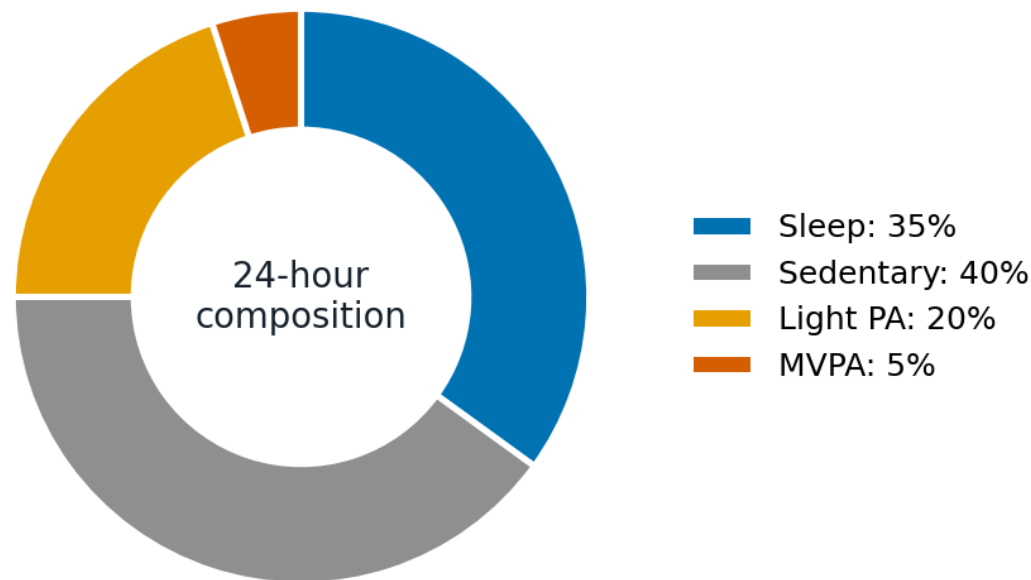


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Dot plot of observed values.

Source: Simulated mock data.

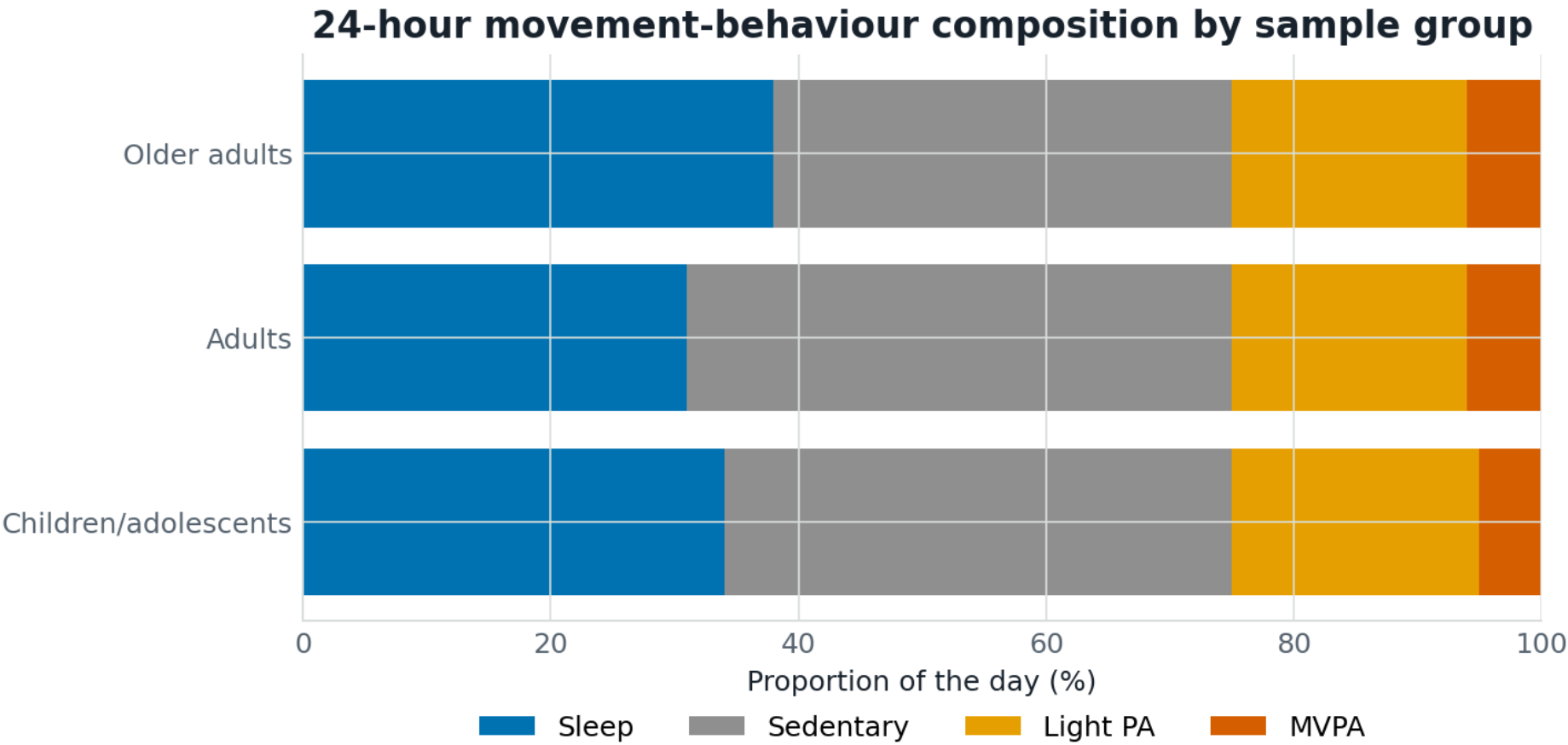
Example 24-hour movement-behaviour composition



Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Pie or doughnut chart.
Source: Simulated mock data.

Simulated example 17. 100% stacked bar chart

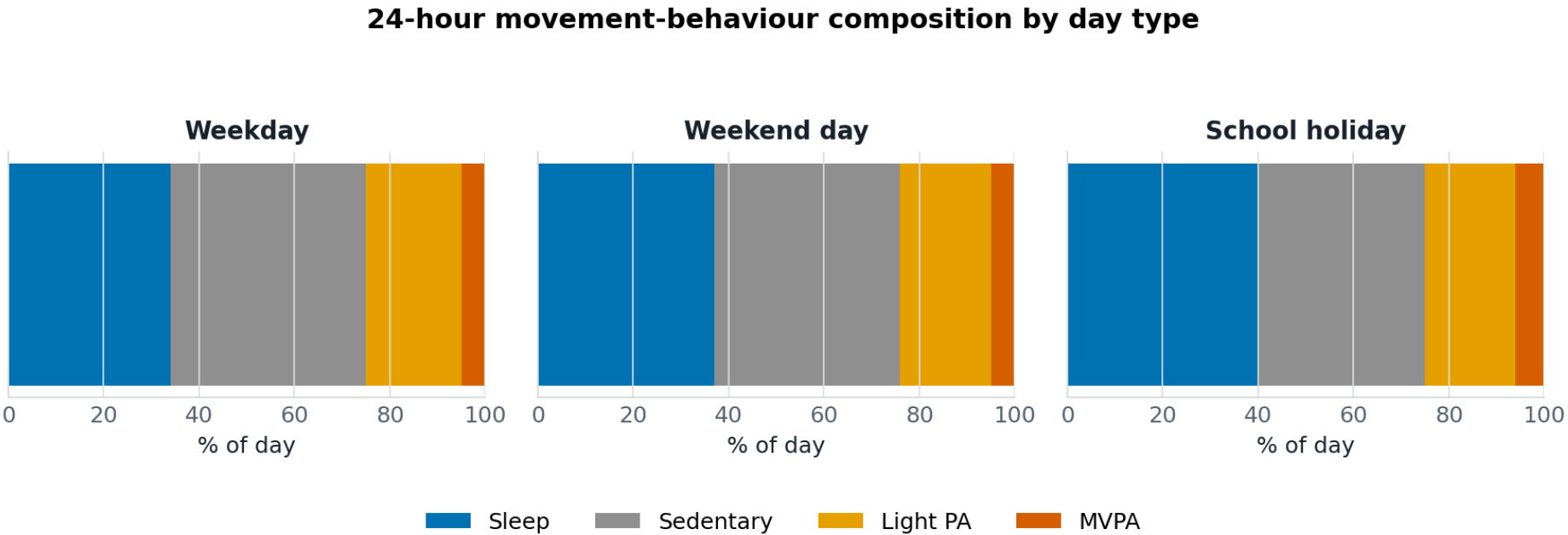


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: 100% stacked bar chart.

Source: Simulated mock data.

Simulated example 18. Small-multiple composition bars



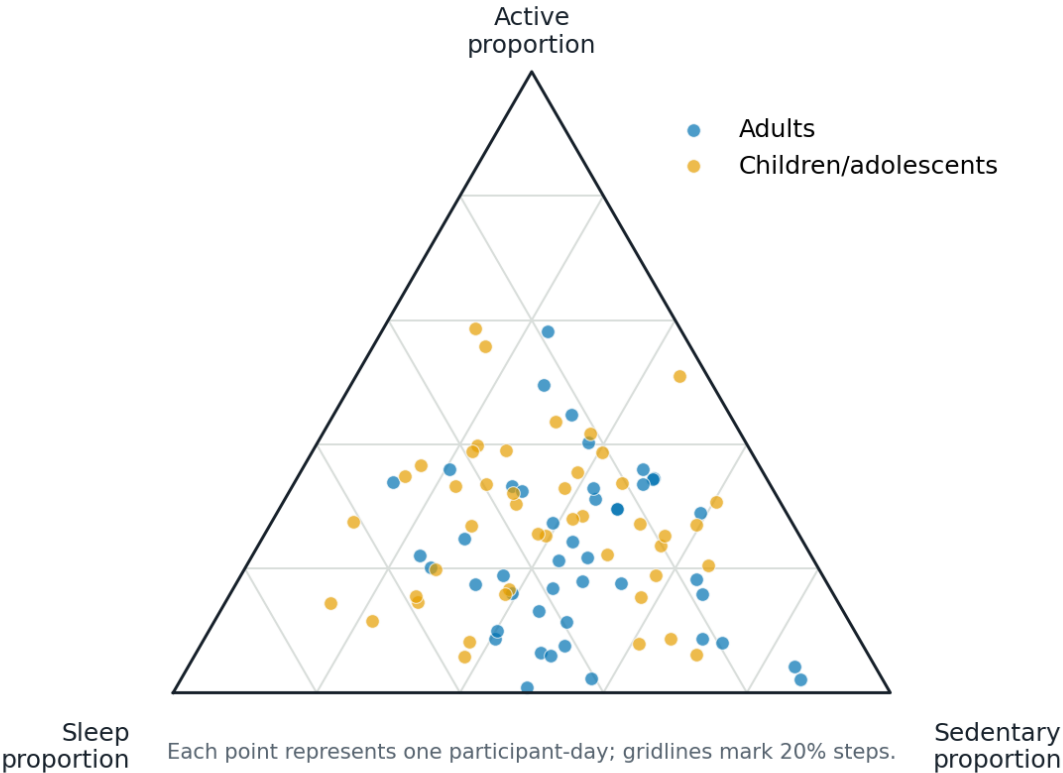
Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Small-multiple composition bars.

Source: Simulated mock data.

Simulated example 19. Ternary plot

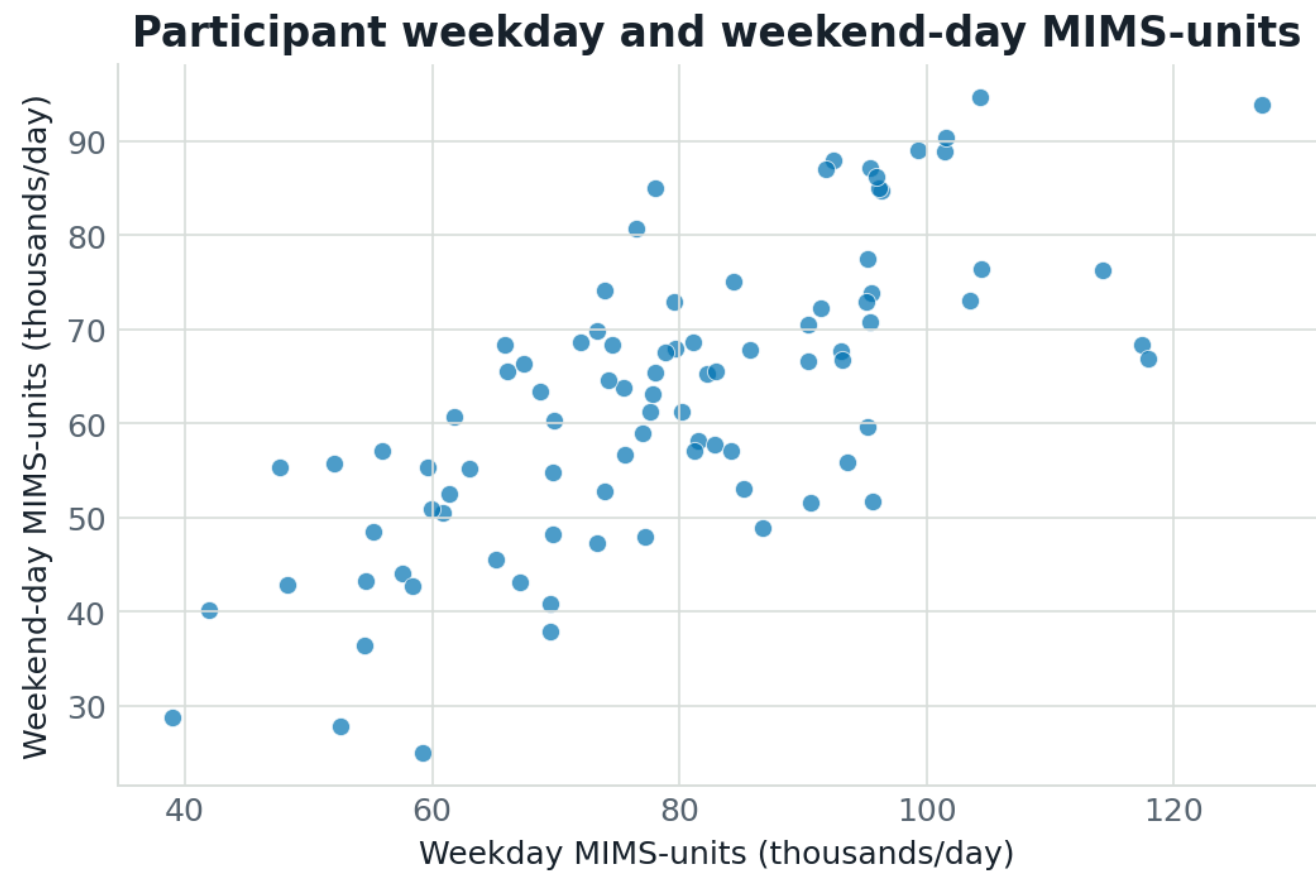
Three-part 24-hour movement-behaviour composition



Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Ternary plot.
Source: Simulated mock data.

Simulated example 20. Scatter plot

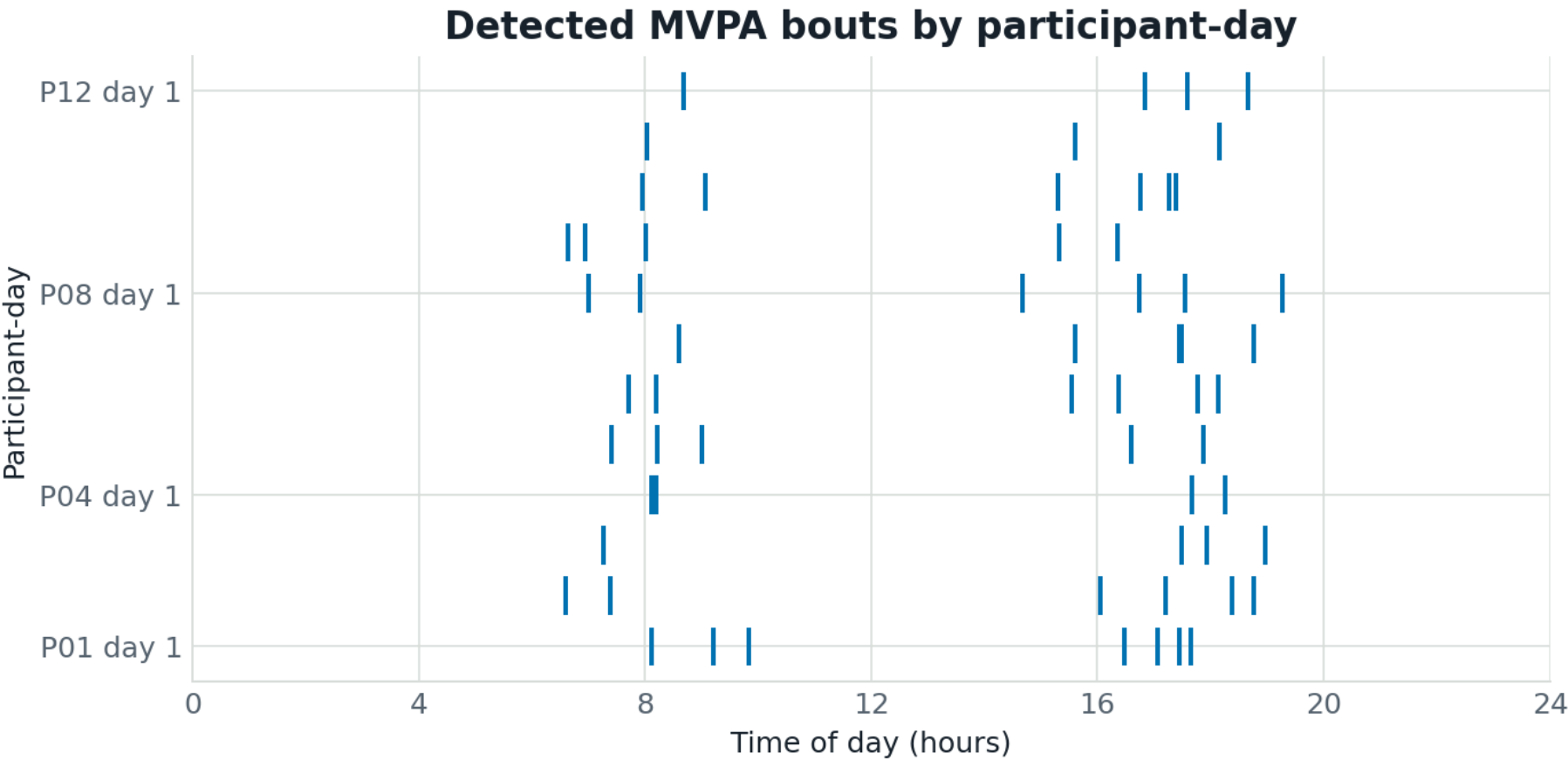


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Scatter plot.

Source: Simulated mock data.

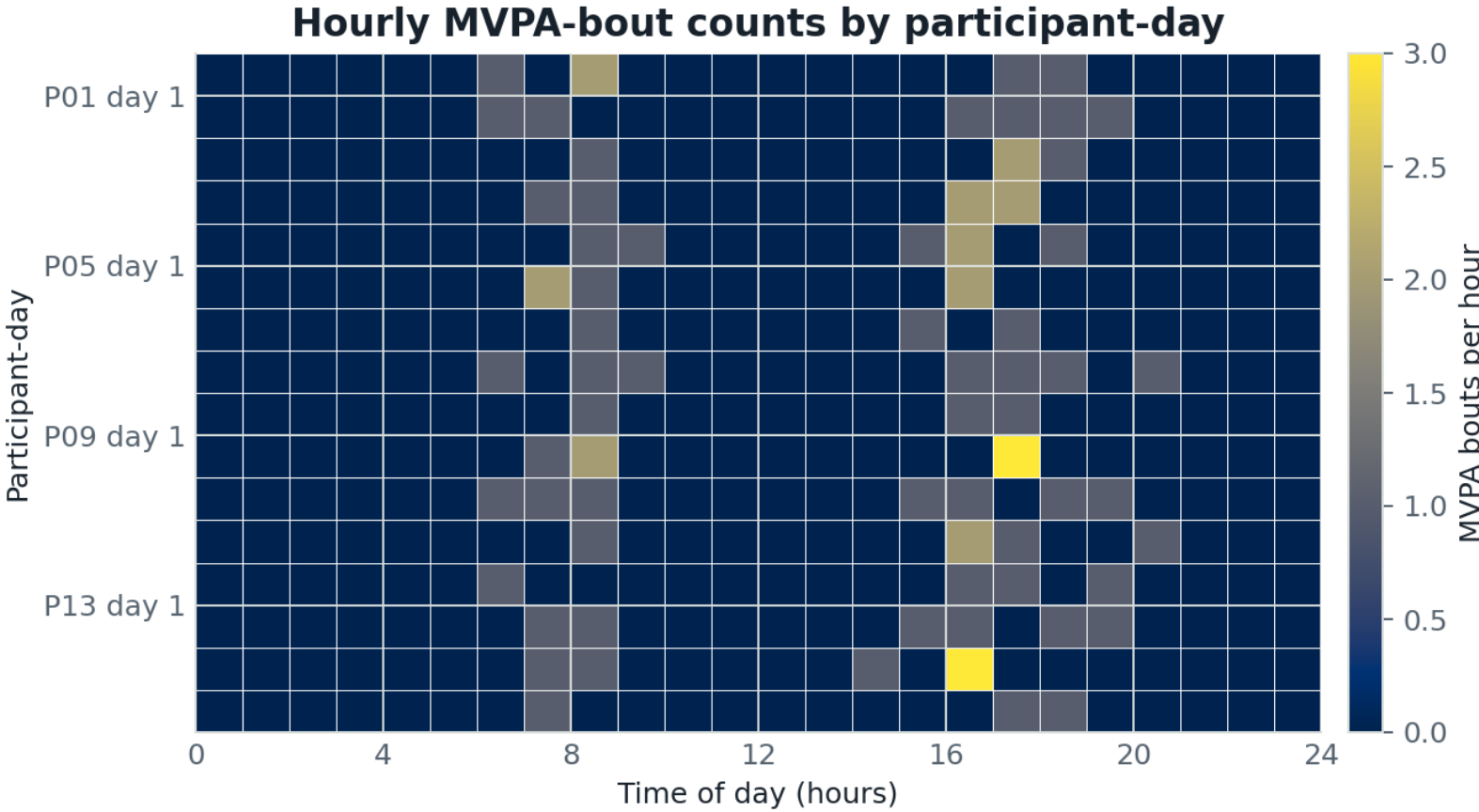
Simulated example 21. Event timeline or raster plot



Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Event timeline or raster plot.
Source: Simulated mock data.

Simulated example 22. Event raster or time-bin heatmap

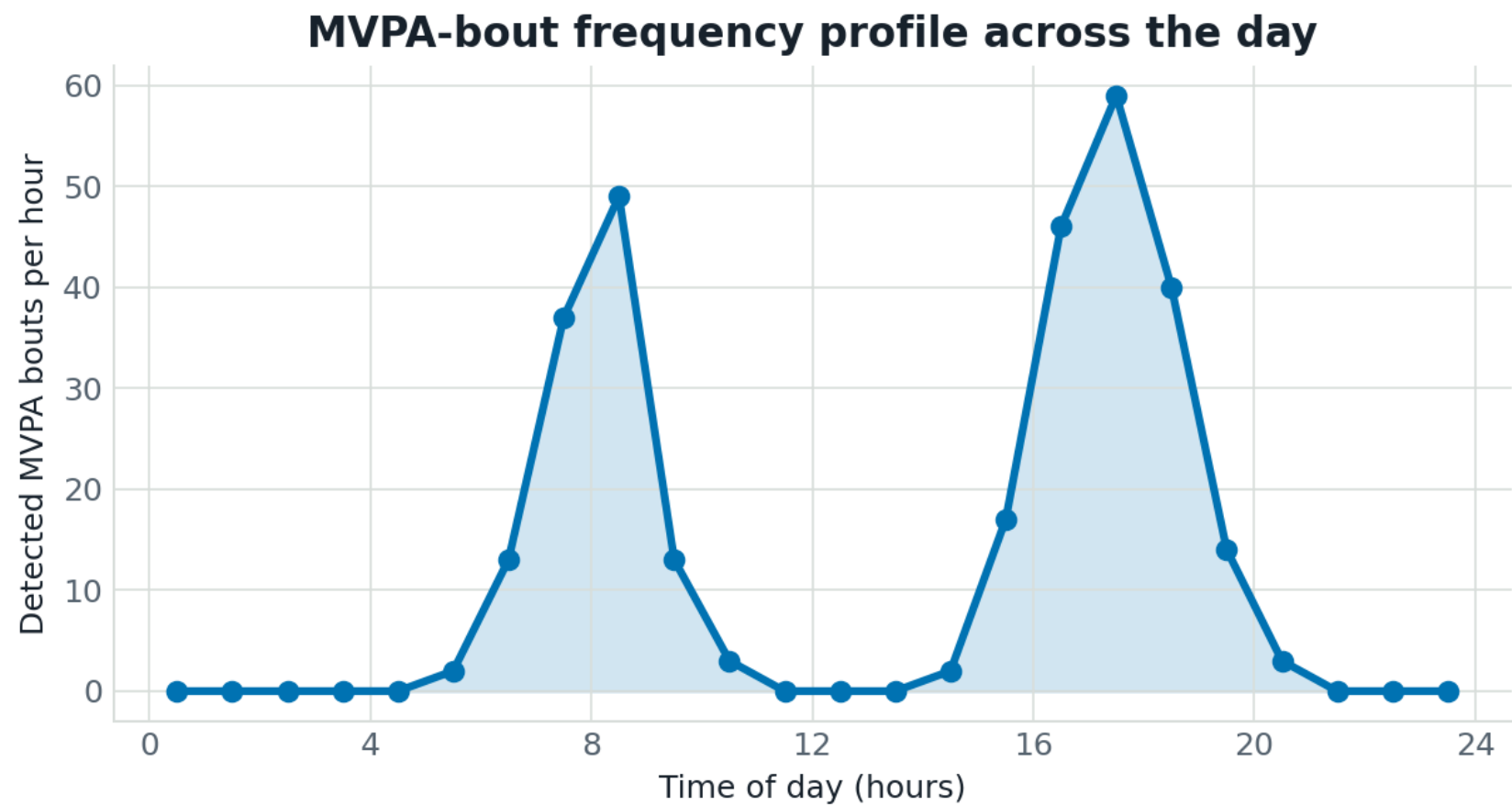


Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Event raster or time-bin heatmap.

Source: Simulated mock data.

Simulated example 23. Event-frequency time profile



Illustrative example generated from simulated mock data

Simulated data example illustrating the visual structure for the recommendation: Event-frequency time profile.
Source: Simulated mock data.